Identification-aided Integrated Evasion and Guidance against Two Non-Cooperative Vehicles via Behavior-Cloning-Enhanced Reinforcement Learning

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Abstract

In three-dimensional terminal flight, an evader may face two Non-Cooperative Vehicles (NCVs) while still required to reach a fixed target under limited overload. Existing evasion guidance is mostly planar and single-NCV oriented, treating evasion, arrival and energy as separable requirements. To address this, this paper proposes an integrated evasion and guidance law enhanced by behavior-cloning-based reinforcement learning. Firstly, a three-dimensional four-body relative motion model is established, and a unified performance index is built on channel-wise zero-effort miss. Then, the maximum principle reduces the coupled terminal conditions to a set of linear algebraic equations, yielding a closed-form feedback guidance law that explicitly integrates evasion against both NCVs and terminal arrival at the target. Furthermore, Kalman filtering is combined with a forgetting-factor recursive least-squares identifier with sparse regularization, which estimates only the non-trivial model coefficients from noisy measurements. Finally, expert trajectories from shooting-based optimization provide behavior cloning pre-training and replay buffer pre-filling, and TD3 fine-tuning schedules the guidance weights online. Monte Carlo simulations show that the evader stays clear of both NCVs while meeting the terminal accuracy requirement. The resulting success rate of simultaneous evasion and target arrival exceeds that of proportional navigation, programmed sinusoidal maneuvering and TD3 without behavior cloning. Under measurement noise, online identification keeps this success rate above that of the same policy without identification.

Keywords: Three-dimensional four-body kinematics; Integrated evasion and guidance; Optimal control; Deep reinforcement learning; Online parameter identification

1 Introduction

When high-speed vehicles perform target-arrival missions in complex dynamic environments, their guidance and control are simultaneously affected by strong nonlinearity, parametric uncertainty, and multiple path and terminal constraints [1]. With the development of multi-vehicle cooperative technologies, a mission vehicle may face multiple approaching non-cooperative vehicles (NCVs) while completing a prescribed arrival mission. Besides prior trajectory planning and cooperative protection [2], terminal active evasive maneuvering remains an important means of ensuring mission safety. Unlike a policy that only seeks a larger single miss distance, an excessively strong or prolonged evasive maneuver consumes limited control authority and reduces the subsequent margin for target correction [3, 4]. This paper focuses on a three-dimensional four-body scenario formed by one evader, two NCVs, and one fixed target. Under a limited normal load factor, the evader must avoid both NCVs while retaining terminal arrival capability at the fixed target. The guidance strategy must therefore coordinate the safe separation from both NCVs, the target-arrival accuracy, and the control effort, rather than treating the three requirements as mutually independent tasks.

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For terminal active evasion, existing studies mainly follow optimal control and differential games. When the pursuer guidance structure is known or can be estimated, optimal control directly poses a one-sided problem in terms of miss distance, energy, and terminal constraints. Early work analyzed optimal evasion on linearized planar kinematics [5]. Later studies considered three-dimensional NCVs using improved proportional navigation together with control constraints [6], the trade-off between a critical safe miss distance and energy consumption [4], and near-optimal feedback evasion against several classes of known linear guidance laws [7]. Another line of work places both players' controls in a zero-sum or linear-quadratic differential game. Turetsky and Shinar compared linear-quadratic and bounded-control game guidance laws for planar one-on-one encounters [8]. These results provide a theoretical baseline for a single evasion encounter. When two evasive encounters and target arrival share the same control input, however, the terminal requirements constrain one another and cannot be decomposed into independent one-on-one problems.

For multi-participant scenarios, existing work has been extended from pure evasion, active protection, and integrated evasion–arrival. Ben-Asher et al. studied planar pure evasion against two pursuers [9]. Yan et al. derived a planar analytical strategy for two successive co-directional pursuers [10]. Zhang et al. periodically recomputed multi-pursuer feasible evasion regions from the current relative state in a three-dimensional orbital scenario [11]. Li et al. used residual reinforcement learning to schedule multi-NCV evasion parameters while accounting for terminal specific energy [3]. The primary objectives of these studies remain safety, miss distance, or energy preservation, which is not the same as an integrated mission that also requires arrival at a fixed target. For integrated missions, Qi et al. split the protected-target problem into two stages, first evading the protective vehicle and then tracking a maneuvering target [12], and Ren et al. coordinated single-NCV evasion and terminal-arrival correction by a hierarchical strategy [13]. Cheng et al. incorporated evasion and target-arrival terminal quantities into differential-game guidance for a planar scenario with multiple mission vehicles, a group of protective vehicles, and a maneuvering target [14]. Further, Hao et al. used reinforcement learning on a planar model to select a parameterized square-wave maneuver for two successive co-directional NCVs while retaining tracking constraints [15]. Shen et al. already considered two NCVs and a terminal target region simultaneously in three-dimensional space and solved the full state–control trajectory by successive SOCP, listing real-time trajectory optimization as subsequent work in the paper [16]. Ren, Wang, and Wu formed a hierarchical framework of predictor–corrector guidance, parameterized evasive correction, and contextual reinforcement learning in a three-dimensional six-degree-of-freedom model [17]. These studies show that three-dimensional integrated modeling of a dual-NCV scenario is not itself an empty area. What still needs to be advanced is to place dual evasion, fixed-target arrival, and the energy trade-off under one optimality condition, so as to obtain a closed-loop analytical structure that does not re-solve a complete trajectory at every guidance cycle.

An analytical structure is only the first step of a closed-loop implementation, because its time-varying model coefficients must still be obtained reliably from online measurement information. Range, range rate, line-of-sight angles, and angular rates provided by relative-state sensors have different measurement mechanisms and error levels [18]. Substituting noisy angular quantities directly into the nonlinear projection relations passes coefficient fluctuations into the guidance command. Existing estimation–guidance studies can recover target states or motion parameters online and feed them back to the guidance loop [19], but that is different from identifying the coupled dual-NCV guidance matrices. The forgetting mechanism in recursive least squares balances tracking of time-varying parameters against steady-state misadjustment [20, 21], and ℓ_1 regularization provides a recursive basis for online estimation of sparse signals [22]. Given a known matrix structure, estimating only the nontrivial elements, suppressing ineffective components caused by weak excitation, and feeding the estimates back to the guidance law is the information link required to move from an ideal analytical solution to an implementation under noisy measurements. Reliable identification answers which model coefficients the guidance computation should use, rather than how the task weights should currently be allocated.

The latter question is an online decision that changes with the encounter situation. The relative importance of the dual-evasion, arrival, and energy terms varies with the geometry and the time-

to-go. Fixed weights cannot cover every phase, whereas solving a nonlinear program at every cycle increases the online computational burden. DDPG provides a deterministic actor–critic framework for continuous-action policy learning [23], and TD3 further mitigates function-approximation error by clipped double Q-learning, delayed policy updates, and target-policy smoothing [24]. Related work has used reinforcement learning for angle-only vehicle encounters [25], for generating guidance commands from the encounter state [26], and for online generation of multi-constraint vehicle trajectories [27], but the output level of those policies is not the guidance-weight scheduling required here. Exploration from scratch in a sparse-reward, multi-constraint environment may also fail to collect enough effective early samples. Demonstration-assisted methods can pre-fill the replay buffer [28] or add a behavior-cloning constraint to the TD3 policy update [29]. In guidance, Hui et al. generated optimal-control samples by the Radau pseudospectral method and learned a single-NCV evasion policy with GAIL–DDPG [30]. This supports converting numerical solutions of the underlying optimal-control problem into task-consistent expert trajectories and then learning a continuous weight-scheduling policy.

In summary, existing studies already cover one-on-one optimal evasion and differential games, pure evasion against multiple pursuers, staged integration with a single NCV, planar parameterized strategies with dual NCVs, full three-dimensional trajectory optimization with two NCVs, and hierarchical intelligent frameworks for multiple NCVs. Moving from these results to a real-time three-dimensional closed-loop implementation with dual NCVs still leaves three linked problems. Dual evasion, fixed-target arrival, and energy consumption must be coupled explicitly in the same optimality condition. The time-varying model coefficients required by the analytical guidance law must be identified online, in a structured way, from noisy measurements. Situation-dependent task weights must be scheduled continuously without trajectory optimization at every cycle, and the underlying optimal-control knowledge should be used to improve policy training. The increment of this paper is therefore the method chain of unified coupled closed-loop analytical guidance, noisy-coefficient identification, and demonstration-enhanced weight scheduling, rather than merely increasing the spatial dimension or the number of participants.

To address the above problems, this paper first establishes a three-dimensional relative-kinematics model for integrated dual-NCV evasion and target arrival, and derives a closed-loop guidance law from optimal control theory. Online parameter identification is then introduced to reduce the effect of measurement errors on the guidance computation. Finally, expert trajectories are generated by the shooting method, and BC-TD3 uses behavior-cloning pre-training, replay-buffer pre-filling, and a decaying behavior-cloning constraint during TD3 fine-tuning to learn the online guidance-parameter policy.

To fill these gaps, this paper proposes an intelligent three-dimensional evasion algorithm for dual non-cooperative vehicles. The main contributions are as follows.

1. A three-dimensional four-body relative-kinematics model is established. The integrated evasion and target-arrival problem is extended from a single non-cooperative vehicle to dual non-cooperative vehicles, and existing planar integrated studies are extended to three-dimensional space.

2. An integrated optimal guidance law is designed that simultaneously accounts for evasion of both non-cooperative vehicles and arrival at the target. A closed-loop analytical form is obtained from the maximum principle. The mutual influence among the evasion tasks and the target-arrival task is taken into account, so that evasion performance and the arrival requirement are balanced and the risk that excessive evasion causes arrival failure is reduced.

3. Considering measurement errors in practical applications, a recursive least-squares identification algorithm with sparse regularization and a forgetting factor is designed. The nontrivial elements of the state-equation matrices are identified online in a structured manner, and the identified results are fed back to the guidance law.

4. Expert trajectories covering a range of initial conditions are generated by the shooting method,

and a BC-TD3 framework is constructed. The expert trajectories are used for behavior-cloning pre-training and replay-buffer pre-filling. During TD3 fine-tuning, a behavior-cloning regularization term is retained in the actor loss and gradually decayed, which transitions the policy from expert imitation to reward-driven optimization and improves training convergence and generalization to unseen scenarios under complex multi-constraint tasks.

The remainder of this paper is organized as follows. Section 2 establishes the three-dimensional four-body relative-kinematics model and formulates the integrated optimization problem. Section 3 derives the integrated optimal guidance law. Section 4 presents the structured sparse FF-RLS online identification method. Section 5 gives expert-trajectory generation and the BC-TD3 framework. Section 6 reports the simulation verification. Section 7 concludes the paper.

2 Problem Formulation

2.1 Nonlinear kinematics

This paper considers an evasion and target-arrival scenario in which two NCVs, D_1 and D_2, are deployed near the target T and approach the evader A cooperatively. For brevity, unless otherwise stated, the subscript $i \in \{1, 2\}$ is used throughout, where $i = 1$ corresponds to NCV D_1 and $i = 2$ to NCV D_2.

The four-body relative geometry among the evader, the target, NCV 1, and NCV 2 is shown in Fig. 1. The frame OXYZ is an Earth-fixed frame with origin at the target T, the X axis in the horizontal plane, and the Y axis pointing vertically upward.

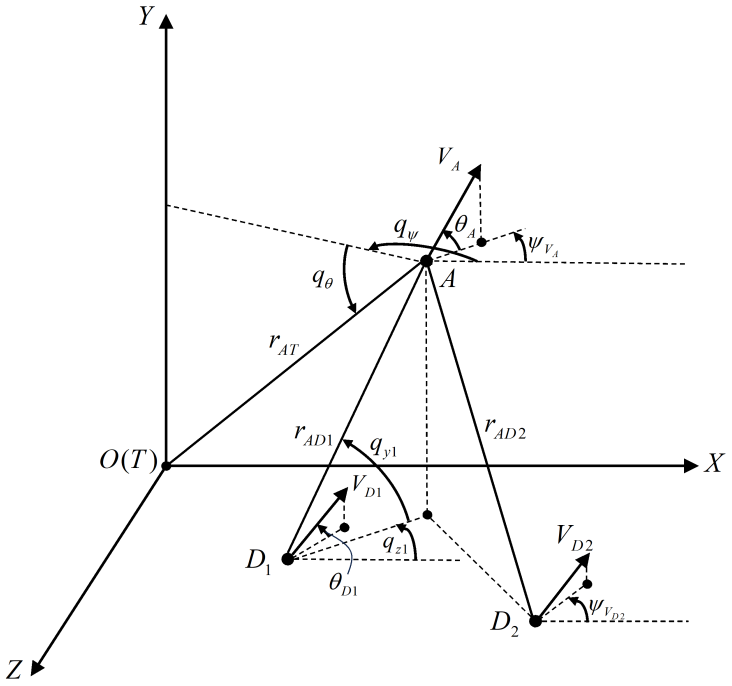


Figure 1: Four-body relative geometry of the evader, the target, NCV 1, and NCV 2.

The symbols for relative position and motion in Fig. 1 are listed in Table 1.

Table 1: Symbols of relative position and motion in Fig. 1.

Symbol	Meaning	Sign in Fig. 1
r_{AT}, r_{ADi}	Ranges from the evader to the target and to the ith NCV	Positive, positive
q_{yi}, q_{zi}	Elevation and azimuth LOS angles of the ith NCV	Positive, positive
q_θ, q_ψ	Elevation and azimuth LOS angles from the evader to the target	Negative, positive
$V_{Di}, \theta_{Di}, \psi_{V_{Di}}$	Speed, flight-path angle, and heading angle of the ith NCV	Positive, positive
$V_A, \theta_A, \psi_{V_A}$	Speed, flight-path angle, and heading angle of the evader	Positive, positive

The relative-motion equations of NCVs D_1 and D_2 with respect to the evader A are

$$\begin{cases} \dot{r}_{ADi} = -V_{Di}\left[\cos\theta_{Di}\cos q_{yi}\cos\left(q_{zi}-\psi_{V_{Di}}\right)+\sin\theta_{Di}\sin q_{yi}\right]+ \\ \quad V_A\left[\cos\theta_A\cos q_{yi}\cos\left(q_{zi}-\psi_{V_A}\right)+\sin\theta_A\sin q_{yi}\right] \\ \dot{q}_{yi}r_{ADi} = V_{Di}\left[\cos\theta_{Di}\sin q_{yi}\cos\left(q_{zi}-\psi_{V_{Di}}\right)-\sin\theta_{Di}\cos q_{yi}\right]- \\ V_A\left[\cos\theta_A\sin q_{yi}\cos\left(q_{zi}-\psi_{V_A}\right)-\sin\theta_A\cos q_{yi}\right] \\ \dot{q}_{zi}r_{ADi}\cos q_{yi} = V_{Di}\cos\theta_{Di}\sin\left(q_{zi}-\psi_{V_{Di}}\right)- \\ \quad V_A\cos\theta_A\sin\left(q_{zi}-\psi_{V_A}\right) \end{cases} \tag{1}$$

The relative-motion equations of the evader and the target are

$$\begin{cases} \dot{r}_{AT} = -V_A\left[\sin q_\theta\sin\theta_A+\cos\left(q_\psi-\psi_{V_A}\right)\cos q_\theta\cos\theta_A\right] \\ \dot{q}_\theta = \frac{V_A\left[\cos\theta_A\sin q_\theta\cos\left(q_\psi-\psi_{V_A}\right)-\sin\theta_A\cos q_\theta\right]}{r_{AT}} \\ \dot{q}_\psi = \frac{V_A\cos\theta_A\sin\left(q_\psi-\psi_{V_A}\right)}{r_{AT}\cos q_\theta} \end{cases} \tag{2}$$

The relative acceleration along the line of sight between each NCV and the evader follows from the kinematic relations:

$$\ddot{r}_{ADi} = r_{ADi}\dot{q}_{yi}^2 + r_{ADi}\dot{q}_{zi}^2\cos^2 q_{yi} + u_{ri} - v_{ri} \tag{3}$$

where u_{ri} and v_{ri} are the projections of the accelerations of the evader and the ith NCV onto the x axis of the NCV–evader line-of-sight frame (denoted S_i). The positive x axis points from the NCV toward the evader.

The relative acceleration along the line of sight between the target and the evader is

$$\ddot{r}_{AT} = r_{AT}\dot{q}_\theta^2 + r_{AT}\dot{q}_\psi^2\cos^2 q_\theta + u_{rT} \tag{4}$$

where u_{rT} is the projection of the evader acceleration onto the x axis of the target–evader line-of-sight frame (denoted S_T). The positive x axis points from the target toward the evader.

For each vehicle, the three-dimensional kinematic equations are

$$\begin{cases} \dot{x}_j = V_j\cos\theta_j\cos\psi_{V_j} \\ \dot{y}_j = V_j\sin\theta_j \\ \dot{z}_j = -V_j\cos\theta_j\sin\psi_{V_j} \\ \dot{\theta}_j = \frac{a_{yj}}{V_j} \\ \dot{\psi}_j = -\frac{a_{zj}}{V_j\cos\theta_j} \end{cases}, j = A, D_1, D_2 \tag{5}$$

where (x_j, y_j, z_j) are the vehicle coordinates in the Earth-fixed frame. Both a_{yj} and a_{zj} are perpendicular to the velocity vector and denote the normal acceleration components along the y and z axes of the flight-path frame. The former lies in the vertical plane containing the velocity vector, and the latter lies in the horizontal xOz plane of the Earth-fixed frame.

From the line-of-sight relative-acceleration relations, the projections u_{ri} and v_{ri} of the evader and the ith NCV accelerations onto the x axis of the line-of-sight frame S_i are related to the normal accelerations a_{yA}, a_{zA} and a_{yDi}, a_{zDi} by a coordinate transformation. Let $\mathbf{L}(q_{yi}, q_{zi})$ be the transformation from the Earth-fixed frame to the line-of-sight frame S_i, and let $\mathbf{L}(\theta, \psi_V)$ be the transformation from the Earth-fixed frame to the flight-path frame. The two projections are then

$$u_{ri} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{L}(q_{yi}, q_{zi}) \mathbf{L}^{-1}(\theta_A, \psi_{V_A}) \begin{bmatrix} 0 \\ a_{yA} \\ a_{zA} \end{bmatrix} \tag{6}$$

$$v_{ri} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{L}(q_{yi}, q_{zi}) \mathbf{L}^{-1}(\theta_{Di}, \psi_{V_{Di}}) \begin{bmatrix} 0 \\ a_{yDi} \\ a_{zDi} \end{bmatrix} \tag{7}$$

The explicit form of the transformation matrices follows the standard definition of three-dimensional coordinate transformations and is omitted here. Likewise, let $\mathbf{L}(-q_\theta, q_\psi - \pi)$ be the transformation from the Earth-fixed frame to the line-of-sight frame S_T, where the line-of-sight angles are the elevation and azimuth corresponding to the positive x axis of S_T (from the target toward the evader). The projection of the evader acceleration onto the x axis of S_T is then

$$u_{rT} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{L}(-q_\theta, q_\psi - \pi) \mathbf{L}^{-1}(\theta_A, \psi_{V_A}) \begin{bmatrix} 0 \\ a_{yA} \\ a_{zA} \end{bmatrix} \tag{8}$$

Assumption 1 (NCV guidance law and performance)**.** *The guidance laws and performance parameters of the two NCVs are known to the evader. Both NCVs employ the classical proportional-navigation (PN) guidance law [31], with acceleration commands*

$$\begin{bmatrix} a_{yDi} \\ a_{zDi} \end{bmatrix} = \begin{bmatrix} K_{Di} V_{Di} \dot{q}_{yi} \\ -K_{Di} V_{Di} \dot{q}_{zi} \cos\theta_{Di} \end{bmatrix} \tag{9}$$

where K_{Di} is the proportional-navigation gain of the ith NCV. The two NCVs have identical performance, that is, $K_{D_1} = K_{D_2} = K_D$ and $V_{D_1} = V_{D_2} = V_D$, and both K_D and V_D remain constant during the engagement.

The desired evasion distances of the two NCVs are taken to be equal, that is, $d_1^* = d_2^* = d^*$.

2.2 Linearized state-space model

Substituting (6), (7), and the associated transformation matrices into (3) yields

$$\begin{aligned} \ddot{r}_{ADi} = r_{ADi} \left(\dot{q}_{yi}^2 + \dot{q}_{zi}^2 \cos^2 q_{yi} \right) - \cos q_{yi} \sin\left(q_{zi} - \psi_{V_A}\right) a_{zA} + \\ a_{yA} \left[-\cos q_{yi} \sin\theta_A \cos\left(q_{zi} - \psi_{V_A}\right) + \sin q_{yi} \cos\theta_A \right] - \\ a_{yDi} \left[-\cos q_{yi} \sin\theta_{Di} \cos\left(q_{zi} - \psi_{V_{Di}}\right) + \sin q_{yi} \cos\theta_{Di} \right] + \\ \cos q_{yi} \sin\left(q_{zi} - \psi_{V_{Di}}\right) a_{zDi} \end{aligned} \tag{10}$$

Likewise, substituting (8) and the associated transformation matrix into (4) yields

$$\begin{aligned} \ddot{r}_{AT} = r_{AT} \left(\dot{q}_\theta^2 + \dot{q}_\psi^2 \cos^2 q_\theta \right) + \left[\sin\theta_A \cos q_\theta \cos\left(q_\psi - \psi_{V_A}\right) - \cos\theta_A \sin q_\theta \right] a_{yA} + \\ \cos q_\theta \sin\left(q_\psi - \psi_{V_A}\right) a_{zA} \end{aligned} \tag{11}$$

Define the state $x_i = \begin{bmatrix} r_{ADi} & \dot{r}_{ADi} \end{bmatrix}^T$ and the control

$$u_A = \begin{bmatrix} a_{yA} & a_{zA} \end{bmatrix}^T$$

. The state-space equation of the interaction between the evader and the ith NCV is then

$$\dot{x}_i(t) = \mathbf{A}_i x_i(t) + \mathbf{B}_i u_A(t) + \mathbf{C}_i \tag{12}$$

where

$$\mathbf{A}_i = \begin{bmatrix} 0 & 1 \\ \dot{q}_{yi}^2 + \dot{q}_{zi}^2 \cos^2 q_{yi} & 0 \end{bmatrix}, \ \mathbf{B}_i = \begin{bmatrix} 0 & 0 \\ B_{1,i} & B_{2,i} \end{bmatrix}, \ \mathbf{C}_i = \begin{bmatrix} 0 \\ C_{1i} \end{bmatrix}$$

with

$$\begin{cases} B_{1,i} = \sin q_{yi} \cos\theta_A - \cos q_{yi} \sin\theta_A \cos\left(q_{zi} - \psi_{V_A}\right) \\ B_{2,i} = -\cos q_{yi} \sin\left(q_{zi} - \psi_{V_A}\right) \\ C_{1i} = \cos q_{yi} \sin\left(q_{zi} - \psi_{V_{Di}}\right) a_{zDi} + \left[\cos q_{yi} \sin\theta_{Di} \cos\left(q_{zi} - \psi_{V_{Di}}\right) - \sin q_{yi} \cos\theta_{Di}\right] a_{yDi} \end{cases} \tag{13}$$

Define the state $x_{AT} = \begin{bmatrix} r_{AT} & \dot{r}_{AT} \end{bmatrix}^T$. The relative state-space equation of the evader and the target is then

$$\dot{x}_{AT}(t) = \mathbf{A}_T x_{AT}(t) + \mathbf{B}_T u_A(t) \tag{14}$$

where

$$\mathbf{A}_T = \begin{bmatrix} 0 & 1 \\ \dot{q}_\theta^2 + \dot{q}_\psi^2 \cos^2 q_\theta & 0 \end{bmatrix}, \ \mathbf{B}_T = \begin{bmatrix} 0 & 0 \\ B_{1,T} & B_{2,T} \end{bmatrix}$$

with

$$\begin{cases} B_{1,T} = \sin\theta_A \cos q_\theta \cos\left(q_\psi - \psi_{V_A}\right) - \cos\theta_A \sin q_\theta \\ B_{2,T} = \cos q_\theta \sin\left(q_\psi - \psi_{V_A}\right) \end{cases} \tag{15}$$

Remark 1. *The system (12)–(15) is linear and time-varying. To obtain an analytical guidance command, each matrix is frozen at its current value and treated as constant in the prediction of a given guidance cycle, and is recomputed from the updated state in the next cycle. This quasi-frozen-coefficient treatment also provides the basis for the online parameter identification in Section 4.*

Assumption 2 (Measurement conditions)**.** *In practice, the matrix elements in (12)–(15) are obtained from the onboard seeker and IMU, rather than from noise-free ideal kinematics. Two classes of measurement noise are considered. The first is angular: the measured line-of-sight angles, line-of-sight rates, and vehicle flight-path and heading angles are each corrupted by zero-mean Gaussian white noise:*

$$\begin{aligned} \tilde{q}_\alpha &= q_\alpha + \nu_{q,\alpha}, \quad \nu_{q,\alpha} \sim \mathcal{N}(0, \sigma_q^2), \\ \tilde{\dot{q}}_\alpha &= \dot{q}_\alpha + \nu_{\dot{q},\alpha}, \quad \nu_{\dot{q},\alpha} \sim \mathcal{N}(0, \sigma_{\dot{q}}^2), \\ \tilde{\theta}_j &= \theta_j + \nu_{\theta,j}, \quad \nu_{\theta,j} \sim \mathcal{N}(0, \sigma_\theta^2), \\ \tilde{\psi}_{V_j} &= \psi_{V_j} + \nu_{\psi,j}, \quad \nu_{\psi,j} \sim \mathcal{N}(0, \sigma_\psi^2), \end{aligned} \tag{16}$$

where $\alpha \in \{y_i, z_i, \theta, \psi\}$ indexes the line-of-sight channels to the ith NCV and to the target, and $j \in \{A, D_1, D_2\}$. The second class is range: the relative range and range rate of each channel are likewise corrupted by zero-mean Gaussian white noise

$$\tilde{r}_c = r_c + \nu_{r,c}, \quad \nu_{r,c} \sim \mathcal{N}(0, \sigma_r^2), \quad \tilde{\dot{r}}_c = \dot{r}_c + \nu_{\dot{r},c}, \quad \nu_{\dot{r},c} \sim \mathcal{N}(0, \sigma_{\dot{r}}^2) \tag{17}$$

where $c \in \{1, 2, T\}$ corresponds to the three range channels r_{AD1}, r_{AD2}, and r_{AT}. A radar seeker measures range from echo delay and range rate from coherent Doppler. Their relative accuracy is limited mainly by signal bandwidth and coherent integration time, whereas angular measurements are limited by beamwidth, target glint, and coupling with vehicle motion, and are typically one to two orders of magnitude less accurate [18]. Accordingly, σ_r and $\sigma_{\dot{r}}$ are taken to be small; their numerical

values are given in Section 6.1. In addition, the relative acceleration $\ddot{r}$ cannot be measured directly and must be estimated online from the noisy $\tilde{r}$ and $\tilde{\dot{r}}$. The normal accelerations a_{yA} and a_{zA} actually applied by the evader are known onboard.

The matrix elements in (13) and (15) are nonlinear functions of four angular measurements. Substituting the measurements directly produces biased and mutually correlated errors that enter the acceleration command through the guidance law. Range measurements are more accurate, and their errors can be further suppressed by filtering. Section 4 therefore builds an online identification framework that recovers the matrix elements from relative-range kinematics, without relying on the noise-sensitive analytical expressions (12)–(15).

2.3 Optimal-problem formulation

To reduce the system order and simplify the solution, the zero-effort miss (ZEM) $z(t)$ is introduced. The ZEM of the evader relative to the ith NCV is defined as

$$z_i(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \Omega\left(t_{fi}, t\right) x_i(t) \tag{18}$$

where the state-transition matrix is

$$\Omega\left(t_{fi}, t\right) = e^{A_i\left(t_{fi}-t\right)} = \begin{bmatrix} \cosh(\dot{Q}_i t_{\mathrm{go,i}}) & \frac{\sinh(\dot{Q}_i t_{\mathrm{go,i}})}{\dot{Q}_i} \\ \dot{Q}_i \sinh(\dot{Q}_i t_{\mathrm{go,i}}) & \cosh(\dot{Q}_i t_{\mathrm{go,i}}) \end{bmatrix} \tag{19}$$

with $\dot{Q}_i = \sqrt{\dot{q}_{yi}^2 + \dot{q}_{zi}^2 \cos^2 q_{yi}}$ the composite line-of-sight rate and $t_{go,i} = t_{fi} - t$ the time-to-go. Using $\dot{\Omega}\left(t_{fi}, t\right) = -\Omega\left(t_{fi}, t\right) A_i$ and differentiating (18) gives

$$\dot{z}_i(t) = \varphi_i(t) b_i^T u_A(t) - \varphi_i(t) v_{ri} \tag{20}$$

where

$$\begin{cases} \varphi_i(t) = \dfrac{e^{\dot{Q}_i t_{go,i}} - e^{-\dot{Q}_i t_{go,i}}}{2\dot{Q}_i} \\ b_i = \begin{bmatrix} B_{1,i} & B_{2,i} \end{bmatrix}^T \end{cases} \tag{21}$$

The derivative of the evader–target ZEM is obtained in the same way:

$$\dot{z}_T(t) = \varphi_T(t) b_T^T u_A(t) \tag{22}$$

where

$$\begin{cases} \varphi_T(t) = \dfrac{e^{\dot{Q}_T t_{go,T}} - e^{-\dot{Q}_T t_{go,T}}}{2\dot{Q}_T} \\ b_T = \begin{bmatrix} B_{1,T} & B_{2,T} \end{bmatrix}^T \end{cases} \tag{23}$$

with $\dot{Q}_T = \sqrt{\dot{q}_\theta^2 + \dot{q}_\psi^2 \cos^2 q_\theta}$ and $t_{go,T} = t_{fT} - t$.

The objective of the intelligent evasion algorithm is to evade both NCVs successfully, minimize the maneuver energy, and avoid excessive maneuvering that would impair subsequent target arrival. The following quadratic performance index is therefore constructed:

$$\min_{u_A} J = \sum\nolimits_{i=1}^{2} \frac{w_i}{2} \left[z_i\left(t_{fi}\right) - d_i^*\right]^2 + \frac{w_E}{2} \int_{t_0}^{t_f} u_A^T u_A d\tau + \frac{w_T}{2} z_T^2\left(t_{fT}\right) \tag{24}$$

where $w_i > 0$ is the weight on the evasion ZEM, $w_E > 0$ is the control-energy weight, $w_T > 0$ is the weight on the target-arrival ZEM, and d_i^* is the desired evasion distance from the ith NCV, usually set slightly larger than the NCV lethal radius. $t_f = \max(t_{f1}, t_{f2})$ is the terminal time of the overall

evasion. To balance the two NCVs, set $w_1 + w_2 = w$ with w a positive constant, and let w_1 and w_2 satisfy

$$\begin{cases} w_1 = w \cdot \dfrac{1}{1+e^{-\gamma}} \\ w_2 = w - w_1 = w \cdot \dfrac{e^{-\gamma}}{1+e^{-\gamma}} \end{cases} \tag{25}$$

Problem 1 (Integrated optimal evasion and guidance)**.** *Consider the dual-NCV proportional-navigation engagement in Assumption 1. Taking the evader normal acceleration $u_A = [a_{yA}, a_{zA}]^T$ as the control, and subject to the three-dimensional four-body kinematics (1)–(5) and the state-space equations (12) ($i = 1, 2$) and (14), together with the command-magnitude limit set by the maximum normal load factor available to the evader, find the optimal control $u_A^*(t)$ that minimizes the performance index (24) expressed by the ZEMs (18) and (22). That is, drive the evasion ZEM toward the desired evasion distance d^* at the engagement instants t_{f1} and t_{f2} of the two NCVs, drive the target-arrival ZEM toward zero at t_{fT}, and minimize the control energy over the whole flight, with the two evasion weights allocated by (25).*

The problem above places evasion of both NCVs, arrival at the target, and control effort under the same optimality condition, rather than treating them by hierarchical decoupling. The integrated guidance law, the guidance-parameter optimization, and the online parameter identification in the remainder of the paper are all developed around its solution.

Solving Problem 1 presents three difficulties.

First, the three tasks are coupled through the same control and the problem is not separable. The control u_A enters the three relative channels simultaneously through the projection coefficients b_1, b_2, and b_T, which vary with the line-of-sight geometry and may change sign. A maneuver that increases the terminal separation from one NCV may reduce the separation from the other and change the ZEM of the target-arrival channel. The three terminal times t_{f1}, t_{f2}, and t_{fT} differ, and the indicator functions further partition the problem in time. Consequently, Problem 1 neither reduces to two independent single-NCV evasion subproblems nor is readily solved by online numerical optimization within the limited time-to-go of the terminal phase. Section 3 therefore applies the maximum principle together with the ZEM order reduction, and reduces the coupling to a 3×3 linear system, which yields a closed-loop analytical guidance command.

Second, the weights in the performance index (24) cannot be fixed a priori. The evasion-weight allocation coefficient γ and the relative magnitudes of w_T and w_E determine the trade-off among evasion, arrival, and energy. Their optimal values depend on the engagement geometry, the time-to-go, and the available load-factor margin. Biasing the weights too far toward evasion keeps the command near the load-factor limit for a long time and removes the subsequent arrival capability, whereas biasing them too far toward arrival leaves insufficient evasion margin. Weight selection is therefore itself a nonlinear optimization problem along the trajectory. Section 5 uses optimal expert trajectories generated by shooting for actor pre-training, replay-buffer pre-filling, and a decaying behavior-cloning constraint during TD3 fine-tuning, after which the trained actor produces the weights online.

Third, the parameters required by the analytical guidance law are not directly available under realistic measurements. The coefficients b_i and b_T, the composite rates $\dot{Q}_i$ and $\dot{Q}_T$, and the NCV maneuver projection v_{ri} are nonlinear functions of the line-of-sight angles, line-of-sight rates, and flight-path angles. Substituting noisy measurements into the analytical expressions (12)–(15) as in Assumption 2 produces biased and mutually correlated errors that enter the guidance command directly, and the relative acceleration cannot be measured directly either. Section 4 therefore constructs a recursive least-squares identifier with sparse regularization and a forgetting factor, which recovers these parameters online from relative-range kinematics and closes the loop with the guidance law.

3 Design of the Integrated Optimal Guidance Law

3.1 Solution of the optimal control problem

The optimal control problem Problem 1 is solved by the maximum principle. The Hamiltonian is

$$H = \lambda_1 \dot{z}_1(t) \cdot I(t \leqslant t_{f1}) + \lambda_2 \dot{z}_2(t) \cdot I(t \leqslant t_{f2}) + \lambda_T \dot{z}_T(t) + \frac{w_E}{2}\left[a_{yA}^2(t) + a_{zA}^2(t)\right] \tag{26}$$

where λ_i and λ_T are the costates of the ZEMs $z_i(t)$ and $z_T(t)$, and $I(t \leqslant t_{fi})$ $(i = 1, 2)$ is an indicator function. The canonical equations and the terminal transversality conditions give the analytical costates

$$\begin{cases} \lambda_i(t) = w_i \left[z_i(t_{fi}) - d_i^*\right], \ i = 1, 2 \\ \lambda_T(t) = w_T z_T(t_{fT}) \end{cases} \tag{27}$$

The stationarity condition $\frac{\partial H}{\partial u_A} = 0$ then yields the open-loop optimal command

$$\begin{aligned} u_A^* = &-\frac{w_1}{w_E}\left[z_1(t_{f1}) - d_1^*\right]\varphi_1(t) b_1 I(t \leqslant t_{f1}) - \frac{w_2}{w_E}\left[z_2(t_{f2}) - d_2^*\right]\varphi_2(t) b_2 I(t \leqslant t_{f2}) - \\ &\frac{w_T}{w_E} z_T(t_{fT})\varphi_T(t) b_T \end{aligned} \tag{28}$$

A closed-loop command requires the terminal ZEMs $z_i(t_{fi})$. Substituting (28) into (20) gives the ZEM derivatives of the evader relative to the two NCVs:

$$\begin{aligned} \dot{z}_1(t) = &-\frac{w_1}{w_E}\left[z_1(t_{f1}) - d_1^*\right]\varphi_1^2(t)\left(B_{1,1}^2 + B_{2,1}^2\right) - \varphi_1(t) v_{r1} - \\ &\frac{w_2}{w_E}\left[z_2(t_{f2}) - d_2^*\right]\varphi_1(t)\varphi_2(t)\left(b_1^T b_2\right) \cdot I(t \leqslant t_{f2}) - \\ &\frac{w_T}{w_E} z_T(t_{fT})\varphi_1(t)\varphi_T(t)\left(b_1^T b_T\right) \ (t \leqslant t_{f1}) \end{aligned} \tag{29}$$

$$\begin{aligned} \dot{z}_2(t) = &-\frac{w_1}{w_E}\left[z_1(t_{f1}) - d_1^*\right]\varphi_1(t)\varphi_2(t)\left(b_1^T b_2\right) \cdot I(t \leqslant t_{f1}) - \\ &\frac{w_2}{w_E}\left[z_2(t_{f2}) - d_2^*\right]\varphi_2^2(t)\left(B_{1,2}^2 + B_{2,2}^2\right) - \varphi_2(t) v_{r2} - \\ &\frac{w_T}{w_E} z_T(t_{fT})\varphi_2(t)\varphi_T(t)\left(b_2^T b_T\right) \ (t \leqslant t_{f2}) \end{aligned} \tag{30}$$

$$\begin{aligned} \dot{z}_T(t) = &-\frac{w_1}{w_E}\left[z_1(t_{f1}) - d_1^*\right]\varphi_T(t)\varphi_1(t)\left(b_1^T b_T\right) \cdot I(t \leqslant t_{f1}) - \\ &\frac{w_2}{w_E}\left[z_2(t_{f2}) - d_2^*\right]\varphi_T(t)\varphi_2(t)\left(b_2^T b_T\right) \cdot I(t \leqslant t_{f2}) - \\ &\frac{w_T}{w_E} z_T(t_{fT})\varphi_T^2(t)\left(B_{1,T}^2 + B_{2,T}^2\right) \ (t \leqslant t_{fT}) \end{aligned} \tag{31}$$

Integrating (29), (30), and (31) from t to t_{f1}, t_{f2}, and t_{fT}, respectively, gives

$$
\begin{aligned}
z_1(t_{f1}) = z_1(t) - \frac{w_1(B_{1,1}^2 + B_{2,1}^2)}{w_E}\Phi_1\left[z_1(t_{f1}) - d_1^*\right] - \\
\frac{w_2\left(b_1^T b_2\right)}{w_E}\Phi_{12}\left[z_2(t_{f2}) - d_2^*\right] - \\
\frac{w_T\left(b_1^T b_T\right)}{w_E}\Phi_{1T} z_T(t_{fT}) - N_1
\end{aligned}
\tag{32}
$$

$$
\begin{aligned}
z_2(t_{f2}) = z_2(t) - \frac{w_1\left(b_1^T b_2\right)}{w_E}\Phi_{12}\left[z_1(t_{f1}) - d_1^*\right] - \\
\frac{w_2(B_{1,2}^2 + B_{2,2}^2)}{w_E}\Phi_2\left[z_2(t_{f2}) - d_2^*\right] - \\
\frac{w_T\left(b_2^T b_T\right)}{w_E}\Phi_{2T} z_T(t_{fT}) - N_2
\end{aligned}
\tag{33}
$$

$$
\begin{aligned}
z_T(t_{fT}) = z_T(t) - \frac{w_1\left(b_1^T b_T\right)}{w_E}\Phi_{1T}\left[z_1(t_{f1}) - d_1^*\right] - \\
\frac{w_2\left(b_2^T b_T\right)}{w_E}\Phi_{2T}\left[z_2(t_{f2}) - d_2^*\right] - \\
\frac{w_T\left(B_{1,T}^2 + B_{2,T}^2\right)}{w_E}\Phi_T z_T(t_{fT})
\end{aligned}
\tag{34}
$$

In the three equations above, Φ_i, N_i (i=1,2), Φ_T, Φ_{12}, Φ_{1T}, and Φ_{2T} are

$$
\begin{cases}
\Phi_i = \displaystyle\int_t^{t_{fi}} \varphi_i^2(\tau)d\tau = \frac{1}{4\dot{Q}_i^2}\left(\frac{e^{2\dot{Q}_i t_{go,i}} - e^{-2\dot{Q}_i t_{go,i}}}{2\dot{Q}_i} - 2t_{go,i}\right),\ i = 1,2 \\
N_i = \displaystyle\int_t^{t_{fi}} \varphi_i(\tau)v_{ri}d\tau = \frac{e^{\dot{Q}_i t_{go,i}} + e^{-\dot{Q}_i t_{go,i}} - 2}{2\dot{Q}_i^2}v_{ri},\ i = 1,2 \\
\Phi_T = \displaystyle\int_t^{t_{fT}} \varphi_T^2(\tau)d\tau = \frac{1}{4\dot{Q}_T^2}\left(\frac{e^{2\dot{Q}_T t_{go,T}} - e^{-2\dot{Q}_T t_{go,T}}}{2\dot{Q}_T} - 2t_{go,T}\right)
\end{cases}
\tag{35}
$$

$$
\begin{aligned}
\Phi_{12} = \int_t^{\min(t_{f1},t_{f2})} \varphi_1(\tau)\varphi_2(\tau)d\tau \\
= \frac{1}{4\dot{Q}_1\dot{Q}_2}\left\{\frac{e^{\dot{Q}_1 t_{go,1} + \dot{Q}_2 t_{go,2}} - e^{\dot{Q}_1 t_{go,1} + \dot{Q}_2 t_{go,2} - (\dot{Q}_1 + \dot{Q}_2)\cdot\min(t_{go,1},t_{go,2})}}{\dot{Q}_1 + \dot{Q}_2}\right. \\
- \frac{e^{\dot{Q}_1 t_{go,1} - \dot{Q}_2 t_{go,2}} - e^{\dot{Q}_1 t_{go,1} - \dot{Q}_2 t_{go,2} - (\dot{Q}_1 - \dot{Q}_2)\cdot\min(t_{go,1},t_{go,2})}}{\dot{Q}_1 - \dot{Q}_2} \\
- \frac{e^{-\dot{Q}_1 t_{go,1} + \dot{Q}_2 t_{go,2} + (\dot{Q}_1 - \dot{Q}_2)\cdot\min(t_{go,1},t_{go,2})} - e^{-\dot{Q}_1 t_{go,1} + \dot{Q}_2 t_{go,2}}}{\dot{Q}_1 - \dot{Q}_2} \\
\left. + \frac{e^{-\dot{Q}_1 t_{go,1} - \dot{Q}_2 t_{go,2} + (\dot{Q}_1 + \dot{Q}_2)\cdot\min(t_{go,1},t_{go,2})} - e^{-\dot{Q}_1 t_{go,1} - \dot{Q}_2 t_{go,2}}}{\dot{Q}_1 + \dot{Q}_2}\right\}
\end{aligned}
\tag{36}
$$

$$
\begin{aligned}
\Phi_{1T} &= \int_t^{\min(t_{f1},t_{fT})} \varphi_1(\tau)\varphi_T(\tau)\,d\tau \\
&= \frac{1}{4\dot{Q}_1\dot{Q}_T}\left\{ \frac{e^{\dot{Q}_1 t_{go,1}+\dot{Q}_T t_{go,T}} - e^{\dot{Q}_1 t_{go,1}+\dot{Q}_T t_{go,T}-(\dot{Q}_1+\dot{Q}_T)\cdot\min(t_{go,1},t_{go,T})}}{\dot{Q}_1+\dot{Q}_T} \right. \\
&\quad - \frac{e^{\dot{Q}_1 t_{go,1}-\dot{Q}_T t_{go,T}} - e^{\dot{Q}_1 t_{go,1}-\dot{Q}_T t_{go,T}-(\dot{Q}_1-\dot{Q}_T)\cdot\min(t_{go,1},t_{go,T})}}{\dot{Q}_1-\dot{Q}_T} \\
&\quad - \frac{e^{-\dot{Q}_1 t_{go,1}+\dot{Q}_T t_{go,T}+(\dot{Q}_1-\dot{Q}_T)\cdot\min(t_{go,1},t_{go,T})} - e^{-\dot{Q}_1 t_{go,1}+\dot{Q}_T t_{go,T}}}{\dot{Q}_1-\dot{Q}_T} \\
&\quad \left. + \frac{e^{-\dot{Q}_1 t_{go,1}-\dot{Q}_T t_{go,T}+(\dot{Q}_1+\dot{Q}_T)\cdot\min(t_{go,1},t_{go,T})} - e^{-\dot{Q}_1 t_{go,1}-\dot{Q}_T t_{go,T}}}{\dot{Q}_1+\dot{Q}_T} \right\}
\end{aligned} \tag{37}
$$

$$
\begin{aligned}
\Phi_{2T} &= \int_t^{\min(t_{f2},t_{fT})} \varphi_2(\tau)\varphi_T(\tau)d\tau \\
&= \frac{1}{4\dot{Q}_2\dot{Q}_T}\left\{ \frac{e^{\dot{Q}_2 t_{go,2}+\dot{Q}_T t_{go,T}} - e^{\dot{Q}_2 t_{go,2}+\dot{Q}_T t_{go,T}-(\dot{Q}_2+\dot{Q}_T)\cdot\min(t_{go,2},t_{go,T})}}{\dot{Q}_2+\dot{Q}_T} \right. \\
&\quad - \frac{e^{\dot{Q}_2 t_{go,2}-\dot{Q}_T t_{go,T}} - e^{\dot{Q}_2 t_{go,2}-\dot{Q}_T t_{go,T}-(\dot{Q}_2-\dot{Q}_T)\cdot\min(t_{go,2},t_{go,T})}}{\dot{Q}_2-\dot{Q}_T} \\
&\quad - \frac{e^{-\dot{Q}_2 t_{go,2}+\dot{Q}_T t_{go,T}+(\dot{Q}_2-\dot{Q}_T)\cdot\min(t_{go,2},t_{go,T})} - e^{-\dot{Q}_2 t_{go,2}+\dot{Q}_T t_{go,T}}}{\dot{Q}_2-\dot{Q}_T} \\
&\quad \left. + \frac{e^{-\dot{Q}_2 t_{go,2}-\dot{Q}_T t_{go,T}+(\dot{Q}_2+\dot{Q}_T)\cdot\min(t_{go,2},t_{go,T})} - e^{-\dot{Q}_2 t_{go,2}-\dot{Q}_T t_{go,T}}}{\dot{Q}_2+\dot{Q}_T} \right\}
\end{aligned} \tag{38}
$$

3.2 Closed-loop guidance command and coupling analysis

Let the unknown ZEM deviations be $\delta z_1 = z_1(t_{f1}) - d_1^*$, $\delta z_2 = z_2(t_{f2}) - d_2^*$, and $\delta z_T = z_T(t_{fT})$. Combining (32), (33), and (34) and rearranging terms gives

$$
\begin{bmatrix} 1+G_{11}\Phi_1 & G_{12}\Phi_{12} & G_{1T}\Phi_{1T} \\ G_{21}\Phi_{12} & 1+G_{22}\Phi_2 & G_{2T}\Phi_{2T} \\ G_{T1}\Phi_{1T} & G_{T2}\Phi_{2T} & 1+G_T\Phi_T \end{bmatrix} \begin{bmatrix} \delta z_1 \\ \delta z_2 \\ \delta z_T \end{bmatrix} = \begin{bmatrix} z_1(t) - N_1 - d_1^* \\ z_2(t) - N_2 - d_2^* \\ z_T(t) \end{bmatrix} \tag{39}
$$

where the self-coupling gains are

$$
G_{11} = \frac{w_1(B_{1,1}^2 + B_{2,1}^2)}{w_E}, \quad G_{22} = \frac{w_2(B_{1,2}^2 + B_{2,2}^2)}{w_E}, \quad G_T = \frac{w_T(B_{1,T}^2 + B_{2,T}^2)}{w_E} \tag{40}
$$

The cross-coupling gains between the NCVs are

$$
G_{12} = \frac{w_2(b_1^T b_2)}{w_E}, \quad G_{21} = \frac{w_1(b_1^T b_2)}{w_E} \tag{41}
$$

The cross-coupling gains between the target and the NCVs are

$$
G_{1T} = \frac{w_T(b_1^T b_T)}{w_E}, \quad G_{2T} = \frac{w_T(b_2^T b_T)}{w_E}, \quad G_{T1} = \frac{w_1(b_1^T b_T)}{w_E}, \quad G_{T2} = \frac{w_2(b_2^T b_T)}{w_E} \tag{42}
$$

Remark 2. *The self-coupling terms G_{11}, G_{22}, and G_T describe the independent effect of the relative motion between each NCV, or the target, and the evader on its own terminal ZEM. The cross-coupling terms G_{12}, G_{21}, G_{1T}, G_{2T}, G_{T1}, and G_{T2} describe the interdependence by which one evader maneuver simultaneously changes the relative motion with respect to both NCVs and the target. This coupling is the essential distinction between the integrated dual-NCV evasion–arrival scenario and*

the single-NCV case, and it is particularly pronounced in a three-dimensional engagement when $b_i^T b_j$ is nontrivial.

Let $\Delta Z_1 = z_1(t) - N_1 - d_1^*$, $\Delta Z_2 = z_2(t) - N_2 - d_2^*$, and $\Delta Z_T = z_T(t)$. Then

$$\begin{bmatrix} 1+G_{11}\Phi_1 & G_{12}\Phi_{12} & G_{1T}\Phi_{1T} \\ G_{21}\Phi_{12} & 1+G_{22}\Phi_2 & G_{2T}\Phi_{2T} \\ G_{T1}\Phi_{1T} & G_{T2}\Phi_{2T} & 1+G_T\Phi_T \end{bmatrix} \begin{bmatrix} \delta z_1 \\ \delta z_2 \\ \delta z_T \end{bmatrix} = \begin{bmatrix} \Delta Z_1 \\ \Delta Z_2 \\ \Delta Z_T \end{bmatrix} \tag{43}$$

Denote the determinant of the coefficient matrix by

$$\Delta_{\text{det}} = \begin{vmatrix} 1+G_{11}\Phi_1 & G_{12}\Phi_{12} & G_{1T}\Phi_{1T} \\ G_{21}\Phi_{12} & 1+G_{22}\Phi_2 & G_{2T}\Phi_{2T} \\ G_{T1}\Phi_{1T} & G_{T2}\Phi_{2T} & 1+G_T\Phi_T \end{vmatrix} \tag{44}$$

$$\Delta_1 = \begin{vmatrix} \Delta Z_1 & G_{12}\Phi_{12} & G_{1T}\Phi_{1T} \\ \Delta Z_2 & 1+G_{22}\Phi_2 & G_{2T}\Phi_{2T} \\ \Delta Z_T & G_{T2}\Phi_{2T} & 1+G_T\Phi_T \end{vmatrix} \tag{45}$$

$$\Delta_2 = \begin{vmatrix} 1+G_{11}\Phi_1 & \Delta Z_1 & G_{1T}\Phi_{1T} \\ G_{21}\Phi_{12} & \Delta Z_2 & G_{2T}\Phi_{2T} \\ G_{T1}\Phi_{1T} & \Delta Z_T & 1+G_T\Phi_T \end{vmatrix} \tag{46}$$

$$\Delta_T = \begin{vmatrix} 1+G_{11}\Phi_1 & G_{12}\Phi_{12} & \Delta Z_1 \\ G_{21}\Phi_{12} & 1+G_{22}\Phi_2 & \Delta Z_2 \\ G_{T1}\Phi_{1T} & G_{T2}\Phi_{2T} & \Delta Z_T \end{vmatrix} \tag{47}$$

Cramer's rule then solves the linear system (43):

$$\delta z_1 = \frac{\Delta_1}{\Delta_{\text{det}}},\ \delta z_2 = \frac{\Delta_2}{\Delta_{\text{det}}},\ \delta z_T = \frac{\Delta_T}{\Delta_{\text{det}}} \tag{48}$$

Substituting (48) into (28) gives the optimal evasion and guidance acceleration command

$$u_A^*(t) = -\frac{1}{\Delta_{\text{det}}} \left\{ \frac{w_1}{w_E}\varphi_1(t)\Delta_1 b_1 I(t \leqslant t_{f1}) \ + \ \frac{w_2}{w_E}\varphi_2(t)\Delta_2 b_2 I(t \leqslant t_{f2}) + \frac{w_T}{w_E}\varphi_T(t)\Delta_T b_T \right\} \tag{49}$$

This optimal command accounts jointly for the evasion requirements of both NCVs, the control-energy consumption, and the effect on target arrival, and thus realizes integrated dual-NCV evasion and target-arrival guidance. As stated in Problem 1, the command (49) is magnitude-limited by the maximum normal load factor available to the evader before it is applied. The effect of this saturation on the evasion and arrival margins is compensated by the online weight scheduling in Section 5. The numerical settings are given in Section 6.1.

Problem 2 (Guidance-parameter optimization)**.** *The closed-loop guidance law* (49) *requires the parameter vector* $\mathbf{p} = [w, \gamma, w_T, w_E]^T$ *to be chosen optimally for each engagement. Finding* $\mathbf{p}^*$ *that meets the desired evasion distance while minimizing energy consumption is a nonlinear optimization problem, which Section 5 solves by the BC-TD3 framework.*

4 Online Parameter Identification

The closed-loop guidance law (49) of Section 3 and its integral gains (35)–(38) depend on the projection coefficients b_i and b_T, the composite line-of-sight rates $\dot{Q}_i$ and $\dot{Q}_T$, and the line-of-sight projection v_{ri} of the NCV maneuver. Under ideal information these quantities can be computed analytically from (12)–(15). Under the measurement conditions of Assumption 2, substituting noisy line-of-sight and flight-path angles directly into the analytical expressions produces biased and mutually correlated errors in the guidance command. During recursive operation, this section therefore

no longer relies on the analytical expressions, which are used only once for filter and parameter initialization. The nontrivial elements of the state-space models (12) and (14) are identified online from relative-range kinematics, and the estimates required by the guidance law are supplied to it.

4.1 Formulation of the parameter-identification problem

By the state-space equations (12) and (14), the second row of each equation contains the nontrivial elements. For an evader–NCV pair, the second-order dynamics extracted from (10) are

$$\ddot{r}_{ADi}(t) = \underbrace{(\dot{q}_{yi}^2 + \dot{q}_{zi}^2 \cos^2 q_{yi})}_{A_{21}^{(i)}} \cdot r_{ADi}(t) + B_{1,i} \cdot a_{yA}(t) + B_{2,i} \cdot a_{zA}(t) + C_{1i}(t) \tag{50}$$

Likewise, for the evader–target pair, (11) gives

$$\ddot{r}_{AT}(t) = \underbrace{(\dot{q}_{\theta}^2 + \dot{q}_{\psi}^2 \cos^2 q_{\theta})}_{A_{21}^{(T)}} \cdot r_{AT}(t) + B_{1,T} \cdot a_{yA}(t) + B_{2,T} \cdot a_{zA}(t) \tag{51}$$

Equations (50)–(51) are linear in the unknown parameters. Comparing (50) with (3) shows that $C_{1i} = -v_{ri}$, the negative of the projection of the ith NCV acceleration onto the line of sight to the evader.

All signals are sampled at the guidance period $t_k = k\Delta t$. The acceleration command computed from the parameter estimate at t_k is held constant on $[t_k, t_{k+1})$. The relative acceleration reconstructed at t_k must therefore be paired with the acceleration actually applied on $[t_{k-1}, t_k)$, not with the command being computed at t_k. The causal regression model of an evader–NCV pair is accordingly defined as

$$y_i(k) = \phi_i^T(k-1)\,\vartheta_i^{[t_{k-1},t_k)} + e_i(k) \tag{52}$$

where $e_i(k)$ is the regression residual of the channel between the evader and the ith NCV, and the remaining quantities are

$$y_i(k) = \hat{\dot{r}}_{ADi}(k|k), \quad \vartheta_i^{[t_{k-1},t_k)} = \begin{bmatrix} A_{21}^{(i)} \\ B_{1,i} \\ B_{2,i} \\ C_{1i} \end{bmatrix}, \quad \phi_i(k-1) = \begin{bmatrix} \hat{r}_{ADi}(k-1|k-1) \\ a_{yA}(k-1) \\ a_{zA}(k-1) \\ 1 \end{bmatrix} \tag{53}$$

For the evader–target pair,

$$y_T(k) = \phi_T^T(k-1)\,\vartheta_T^{[t_{k-1},t_k)} + e_T(k) \tag{54}$$

where $e_T(k)$ is the regression residual of the evader–target channel, and the remaining quantities are

$$y_T(k) = \hat{\dot{r}}_{AT}(k|k), \quad \vartheta_T^{[t_{k-1},t_k)} = \begin{bmatrix} A_{21}^{(T)} \\ B_{1,T} \\ B_{2,T} \end{bmatrix}, \quad \phi_T(k-1) = \begin{bmatrix} \hat{r}_{AT}(k-1|k-1) \\ a_{yA}(k-1) \\ a_{zA}(k-1) \end{bmatrix} \tag{55}$$

In (53) and (55), $a_{yA}(k-1)$ and $a_{zA}(k-1)$ are the saturated accelerations actually applied on $[t_{k-1}, t_k)$. The quantity $\hat{\dot{r}}(k|k)$ is the posterior estimate of the relative acceleration using measurements up to t_k, and $\hat{r}(k-1|k-1)$ in the regressor is the posterior range estimate at the previous instant. Both are provided by the Kalman filter in Section 4.2. The vectors $\vartheta_i^{[t_{k-1},t_k)}$ and $\vartheta_T^{[t_{k-1},t_k)}$ in (52) and (54) are equivalent parameters on $[t_{k-1}, t_k)$. The physical parameters they represent vary continuously with the line-of-sight geometry, but they are treated as constant within a single regression equation. Equation (56) then fits all data in the forgetting window by the same constant associated with the current interval, and the slow variation of the parameters is tracked by

the forgetting factor ρ. The estimates $\hat{\vartheta}_i(k)$ and $\hat{\vartheta}_T(k)$ obtained at t_k correspond to $\vartheta_i^{[t_{k-1},t_k)}$ and $\vartheta_T^{[t_{k-1},t_k)}$, and are the latest estimates available. The residuals $e_i(k)$ and $e_T(k)$ consist mainly of the relative-acceleration estimation error and the modeling error incurred by representing time-varying parameters in the forgetting window by a single constant.

Structured identification uses the known sparsity pattern of the state matrices: the $(1,1)$, $(1,2)$, and $(2,2)$ entries of $\mathbf{A}_i$ and $\mathbf{A}_T$ are fixed at 0, 1, and 0, and the first rows of $\mathbf{B}_i$ and $\mathbf{B}_T$ are zero. Only the nontrivial elements are estimated, which reduces the parameter dimension and preserves physical consistency. These elements are summarized in Table 2.

Table 2: Nontrivial elements to be identified.

Equation	Matrix	Element	Physical meaning
(12)	$\mathbf{A}_i$	$A_{21}^{(i)} = \dot{q}_{yi}^2 + \dot{q}_{zi}^2 \cos^2 q_{yi}$	Line-of-sight rate coupling
(12)	$\mathbf{B}_i$	$B_{1,i}, B_{2,i}$	Evader acceleration projection coefficients (NCV direction)
(12)	$\mathbf{C}_i$	C_{1i}	NCV maneuver-acceleration bias
(14)	$\mathbf{A}_T$	$A_{21}^{(T)} = \dot{q}_\theta^2 + \dot{q}_\psi^2 \cos^2 q_\theta$	Target–evader line-of-sight rate coupling
(14)	$\mathbf{B}_T$	$B_{1,T}, B_{2,T}$	Evader acceleration projection coefficients (target direction)

Under the measurement conditions of Assumption 2, $\ddot{r}$ must be estimated online from the noisy $\tilde{r}$ and $\tilde{\dot{r}}$. The regression residuals $e_i(k)$ and $e_T(k)$ are therefore nonzero, and some free parameters, especially the projection coefficients $B_{1,i}$, $B_{2,i}$, C_{1i}, $B_{1,T}$, and $B_{2,T}$, may be weakly excited in certain geometries. To improve identification stability under weak excitation and to suppress spurious parameter components caused by relative-acceleration estimation error, the diagonal scaling in (57) is used, and an ℓ_1 (Lasso) penalty is added to the forgetting-factor least-squares cost in the normalized domain [32]. The superscript "sc" denotes parameters after this scaling. For an evader–NCV pair $(i = 1, 2)$,

$$J_{i,k}\left(\vartheta_i^{\mathrm{sc},[t_{k-1},t_k)}\right) = \sum_{n=1}^{k} \rho^{k-n}\left[y_i(n) - \bar{\phi}_i^T(n-1)\vartheta_i^{\mathrm{sc},[t_{k-1},t_k)}\right]^2 + \mu \sum_{j=2}^{4} \left|[\vartheta_i^{\mathrm{sc},[t_{k-1},t_k)}]_j\right| \tag{56}$$

The same cost $J_{T,k}\left(\vartheta_T^{\mathrm{sc},[t_{k-1},t_k)}\right)$ is constructed for the evader–target pair, with the ℓ_1 sum taken over $j = 2, 3$. Here k is the current recursion index and $n = 1, \ldots, k$ indexes the past samples; $\rho \in (0, 1]$ is the forgetting factor and $\mu \geqslant 0$ is the sparsity weight. In both costs the ℓ_1 penalty acts only on the projection coefficients and the bias components. The entry A_{21} is a nonnegative geometric parameter determined by the sum of squared line-of-sight rates and enters the guidance law through $\dot{Q} = \sqrt{A_{21}}$, so it is not soft-thresholded as a sparse component. Its nonnegativity is enforced by the projection (67). The normalizing transformation is

$$\bar{\phi} = M^{-1}\phi, \quad \vartheta^{\mathrm{sc},[t_{k-1},t_k)} = M\vartheta^{[t_{k-1},t_k)}, \quad M_i = \mathrm{diag}(r_0, a_0, a_0, 1), \quad M_T = \mathrm{diag}(r_0, a_0, a_0) \tag{57}$$

where r_0 and a_0 are the nominal range and acceleration scales. The transformation leaves the predicted measurement unchanged, that is,

$$\bar{\phi}^T \vartheta^{\mathrm{sc},[t_{k-1},t_k)} = \phi^T \vartheta^{[t_{k-1},t_k)}.$$

The ℓ_1 term in (56) penalizes the projection coefficients and the bias components, starting from the second component, in this normalized domain. In the recursion below the superscript "sc" is omitted for brevity. All variables are understood to be normalized, and $\hat{\vartheta}_i(k)$ and $\hat{\vartheta}_T(k)$ still denote the step-k estimates of the equivalent parameters on the current interval.

Problem 3 (Structured sparse parameter identification)**.** *Given the seeker measurements of relative range and range rate in Assumption 2 and the acceleration actually applied by the evader, and subject to the known sparsity pattern of the state matrices, take the forgetting-factor ℓ_1-regularized criterion (56) as the design criterion. At each guidance period Δt, using only information up to t_k, recursively estimate $\vartheta_i^{[t_{k-1},t_k)}$ $(i=1,2)$ and $\vartheta_T^{[t_{k-1},t_k)}$ by a proximal (soft-threshold) approximation of this criterion, and thereby provide online estimates of $\mathbf{A}_i$, $\mathbf{B}_i$, $\mathbf{C}_i$, $\mathbf{A}_T$, and $\mathbf{B}_T$ to the guidance law.*

4.2 Kalman estimation of relative acceleration and range

Under the measurement conditions of Assumption 2, the relative acceleration $\ddot{r}$ cannot be measured directly, and numerical differentiation of the noisy range rate amplifies the measurement noise by $1/\Delta t$, which is a factor of ten when $\Delta t = 0.1\text{s}$. An independent three-state constant-acceleration Kalman filter is therefore assigned to each of the two NCV range channels $i \in \{1,2\}$ and to the target range channel T. In what follows, c denotes any one of these range channels. Equations (12) and (14) are not used as the filter model, because the nontrivial elements $A_{21}^{(i)}, B_{1,i}, B_{2,i}, C_{1i}$ of an NCV channel and $A_{21}^{(T)}, B_{1,T}, B_{2,T}$ of the target channel are exactly the quantities to be identified. Building the filter from their estimates would create a circular dependence between identification and filtering. A constant-acceleration model independent of the line-of-sight geometry, with the relative jerk treated as white noise, makes the reconstruction of $\ddot{r}$ independent of the parameters being estimated. The state and measurement are

$$\xi_c(k) = [r_c(k), \dot{r}_c(k), \ddot{r}_c(k)]^T, \quad z_{\xi,c}(k) = [\tilde{r}_c(k), \tilde{\dot{r}}_c(k)]^T \tag{58}$$

The discrete process and measurement equations of each channel are

$$\xi_c(k) = \mathbf{F}_\xi \xi_c(k-1) + w_c(k-1), \quad z_{\xi,c}(k) = \mathbf{H}_\xi \xi_c(k) + \nu_c(k) \tag{59}$$

where the state-transition and measurement matrices are

$$\mathbf{F}_\xi = \begin{bmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H}_\xi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \tag{60}$$

The process noise $w_c \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_\xi)$ and the measurement noise $\nu_c \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_\xi)$ are uncorrelated. Modeling the relative jerk as continuous white noise of intensity σ_j^2, the discrete-equivalent process covariance and the measurement covariance determined by (17) are [33]

$$\mathbf{Q}_\xi = \sigma_j^2 \begin{bmatrix} \Delta t^5/20 & \Delta t^4/8 & \Delta t^3/6 \\ \Delta t^4/8 & \Delta t^3/3 & \Delta t^2/2 \\ \Delta t^3/6 & \Delta t^2/2 & \Delta t \end{bmatrix}, \quad \mathbf{R}_\xi = \operatorname{diag}(\sigma_r^2, \sigma_{\dot{r}}^2) \tag{61}$$

Within each guidance cycle the standard Kalman recursion first predicts and then updates [33]:

$$\begin{cases} \hat{\xi}_c^-(k) = \mathbf{F}_\xi \hat{\xi}_c(k-1|k-1), \\ \mathbf{P}_c^-(k) = \mathbf{F}_\xi \mathbf{P}_c(k-1|k-1)\mathbf{F}_\xi^T + \mathbf{Q}_\xi, \\ \mathbf{K}_{\xi,c}(k) = \mathbf{P}_c^-(k)\mathbf{H}_\xi^T \left[\mathbf{H}_\xi \mathbf{P}_c^-(k)\mathbf{H}_\xi^T + \mathbf{R}_\xi\right]^{-1}, \\ \hat{\xi}_c(k|k) = \hat{\xi}_c^-(k) + \mathbf{K}_{\xi,c}(k)\left[z_{\xi,c}(k) - \mathbf{H}_\xi \hat{\xi}_c^-(k)\right], \\ \mathbf{P}_c(k|k) = \left[\mathbf{I} - \mathbf{K}_{\xi,c}(k)\mathbf{H}_\xi\right]\mathbf{P}_c^-(k). \end{cases} \tag{62}$$

The regression output $y_c(k)$ is the third component of $\hat{\xi}_c(k|k)$, and the range entry of the regressor is the first component of $\hat{\xi}_c(k-1|k-1)$.

The filter is initialized by $\hat{\xi}_c(0|0) = [\tilde{r}_c(0), \tilde{\dot{r}}_c(0), \ddot{r}_{c,0}]^T$, where $\ddot{r}_{c,0}$ is obtained by substituting the initial measurements into the analytical expressions (10) and (11). The NCV acceleration

required by an NCV channel is given by the PN model (9). The initial covariance is $\mathbf{P}_c(0|0) = \mathrm{diag}(\sigma_r^2, \sigma_{\dot{r}}^2, \sigma_{a0}^2)$, where σ_{a0} is chosen on the order of the sum of the normal load factors available to the evader and the NCVs. The jerk intensity σ_j sets the trade-off between tracking bandwidth and noise suppression. If σ_j is too small, the lag of $\hat{\dot{r}}$ grows and biases the estimate when C_{1i} changes rapidly with the NCV proportional-navigation acceleration in the terminal phase. If σ_j is too large, more measurement noise enters y_c and then the parameter estimate through (64). The values of σ_j, σ_r, and $\sigma_{\dot{r}}$ are given together in Section 6.1.

4.3 Sparse FF-RLS with an ℓ_1 proximal update

For Problem 3, this subsection constructs a sparse FF-RLS estimator with an ℓ_1 proximal update. In each guidance cycle, a forgetting-factor recursive least-squares quadratic update is performed first [32, 34]. An element-wise soft threshold then realizes the sparse shrinkage corresponding to the ℓ_1 penalty in (56), and the estimate of A_{21} is projected onto its physically feasible set. This improves tracking of the time-varying parameters under weak excitation and noisy residuals. The soft-threshold step is a separable approximation of the proximal operator of the ℓ_1 term in (56). It suppresses small ineffective parameter components caused by noise and weak excitation while preserving the recursive form and the online computational cost. The forgetting factor ρ discounts old data exponentially, so that the estimator can follow the continuing change of the line-of-sight geometry during the engagement. For an evader–NCV pair ($i = 1, 2$), the quadratic update is

$$\mathbf{K}_i(k) = \frac{\mathbf{P}_i(k-1)\,\phi_i(k-1)}{\rho + \phi_i^T(k-1)\mathbf{P}_i(k-1)\,\phi_i(k-1)} \tag{63}$$

$$\bar{\vartheta}_i(k) = \hat{\vartheta}_i(k-1) + \mathbf{K}_i(k)[y_i(k) - \phi_i^T(k-1)\,\hat{\vartheta}_i(k-1)] \tag{64}$$

$$\mathbf{P}_i(k) = \frac{1}{\rho}[\mathbf{I} - \mathbf{K}_i(k)\,\phi_i^T(k-1)]\,\mathbf{P}_i(k-1) \tag{65}$$

where $\mathbf{K}_i(k)$ is the gain, $\mathbf{P}_i(k)$ is the estimation-error covariance, and $\bar{\vartheta}_i(k)$ is the intermediate estimate from the quadratic update. The update of $\bar{\vartheta}_T(k)$ for the evader–target pair has the same form, with every subscript i replaced by T. After the RLS update, one element-wise soft threshold is applied to the projection coefficients and the bias components, which realizes the sparse shrinkage of the ℓ_1 penalty:

$$\left[\hat{\vartheta}_i(k)\right]_j = \begin{cases} \left[\bar{\vartheta}_i(k)\right]_j, & j = 1 \\ \left[\mathcal{S}_\eta\left(\bar{\vartheta}_i(k)\right)\right]_j, & j = 2, 3, \ldots \end{cases} \tag{66}$$

where $[\mathcal{S}_\eta(v)]_j = \mathrm{sign}(v_j)\max(|v_j| - \eta, 0)$ and $\eta \geqslant 0$ is the soft threshold in the normalized parameter domain, corresponding to the sparsity weight μ in (56) and calibrated by offline simulation. The index $j = 1$ corresponds to A_{21} and is not shrunk, while $j \geqslant 2$ corresponds to the projection coefficients and the bias components.

The same operation is applied to $\bar{\vartheta}_T(k)$. The vector $\hat{\vartheta}_i(k)$ is the final estimate at step k. In the next cycle, $\hat{\vartheta}_i(k-1)$ in (64) takes the value after soft thresholding and the projection below. Because A_{21} is a sum of squares and enters the guidance law through $\dot{Q} = \sqrt{A_{21}}$, its estimate is projected onto the physically feasible set:

$$\hat{A}_{21}^{(i)}(k) \leftarrow \max(\hat{A}_{21}^{(i)}(k), 0), \quad i \in \{1, 2\}, \qquad \hat{A}_{21}^{(T)}(k) \leftarrow \max(\hat{A}_{21}^{(T)}(k), 0) \tag{67}$$

The parameter estimates are initialized from the analytical expressions (12)–(15). The initial values are computed from the initial measurements, scaled by M, and then enter the recursion. At the same time, $\mathbf{P}_i(0) = \delta^{-1}\mathbf{I}$ and $\mathbf{P}_T(0) = \delta^{-1}\mathbf{I}$ with $\delta \ll 1$. For an NCV channel, the initial value of C_{1i} uses the NCV acceleration given by the PN model (9).

Finally, the identified parameters are reassembled into the state-space matrices:

$$\hat{\mathbf{A}}_i = \begin{bmatrix} 0 & 1 \\ \hat{A}_{21}^{(i)} & 0 \end{bmatrix}, \quad \hat{\mathbf{B}}_i = \begin{bmatrix} 0 & 0 \\ \hat{B}_{1,i} & \hat{B}_{2,i} \end{bmatrix}, \quad \hat{\mathbf{C}}_i = \begin{bmatrix} 0 \\ \hat{C}_{1i} \end{bmatrix} \tag{68}$$

$$\hat{\mathbf{A}}_T = \begin{bmatrix} 0 & 1 \\ \hat{A}_{21}^{(T)} & 0 \end{bmatrix}, \quad \hat{\mathbf{B}}_T = \begin{bmatrix} 0 & 0 \\ \hat{B}_{1,T} & \hat{B}_{2,T} \end{bmatrix} \tag{69}$$

Every estimate in (68)–(71) has been transformed back to physical units by $\hat{\vartheta}(k) = M^{-1}\hat{\vartheta}^{\mathrm{sc}}(k)$. The quantities required by the guidance law for the ith NCV are

$$\hat{\hat{Q}}_i = \sqrt{\hat{A}_{21}^{(i)}}, \quad \hat{b}_i = \left[\hat{B}_{1,i}, \hat{B}_{2,i}\right]^T, \quad \hat{v}_{ri} = -\hat{C}_{1i} \tag{70}$$

The quantities required for the target are

$$\hat{\hat{Q}}_T = \sqrt{\hat{A}_{21}^{(T)}}, \quad \hat{b}_T = \left[\hat{B}_{1,T}, \hat{B}_{2,T}\right]^T \tag{71}$$

These estimates replace the corresponding analytically computed quantities in the optimal guidance law (49) and the integral gains (35)–(38), which forms a closed-loop identification–guidance architecture.

Remark 3. *The accuracy of the identifier is affected by the following factors. When the forgetting window is short, r varies slowly inside the window and is difficult to distinguish from the constant regressor entry. The accelerations a_{yA} and a_{zA} are produced by the closed-loop guidance law, so whether each parameter is sufficiently excited depends on how rich the maneuver is during the engagement. The regression output $\hat{\ddot{r}}$ is estimated by a constant-acceleration Kalman filter. That model can still lag where $\ddot{r}$ changes rapidly, and the trade-off between lag and noise suppression is set by σ_j, so some bias remains in the parameter estimates. In addition, C_{1i} varies rapidly with the NCV proportional-navigation acceleration in the terminal phase, and the approximation of treating it as constant over one guidance cycle becomes weaker.*

5 Guidance-Parameter Optimization Based on BC-TD3

This section develops a BC-TD3 framework for guidance-parameter optimization to solve Problem 2. First, using the structure of the closed-loop guidance law (49), the continuous-time parameter optimization problem is transcribed by single shooting into a finite-dimensional nonlinear program (NLP). Expert trajectories that satisfy the minimum evasion distance and the target-arrival accuracy are then generated offline under randomly sampled initial conditions. A BC-TD3 network is constructed next. The actor is pre-trained by behavior cloning on the expert demonstrations, and the demonstrations are used to pre-fill the replay buffer. During subsequent TD3 fine-tuning, a decaying behavior-cloning regularization term is retained in the actor loss, which gradually transfers the policy from expert imitation to reward-driven optimization.

5.1 Expert-trajectory generation

In the training scenarios, the simulated initial positions of the NCVs coincide with the target, that is, with the origin of the frame OXYZ. From the geometry, the lead angles of an NCV velocity are

$$\begin{cases} \eta_{yDi} = \theta_{Di} - q_{yi} \\ \eta_{zDi} = \psi_{V_{Di}} - q_{zi} \end{cases}, \ i = 1, 2 \tag{72}$$

The lead angles of the evader velocity relative to the ith NCV are

$$\begin{cases} \eta_{yAi} = \theta_A + q_{yi} \\ \eta_{zAi} = \psi_{V_A} - q_{zi} - \pi \end{cases} \tag{73}$$

Taking the main factors that affect evasion effectiveness into account, the initial angular state of the evader relative to the ith NCV is defined as $X_i = [q_{yi0}, q_{zi0}, \eta_{yAi0}, \eta_{zAi0}, \eta_{yDi0}, \eta_{zDi0}]$ $(i = 1, 2)$. In the training samples, when the evader enters an evasive state against one NCV, the difference between the two NCV–evader ranges is set not to exceed 3 km. Let the random variable $\beta \in [-1.5, 1.5]$km be an adjustment variable and r_{AD0} the midpoint, with $r_{AD10} = r_{AD0} + \beta$ and $r_{AD20} = r_{AD0} - \beta$. The 14-dimensional initial state $X = [X_1, X_2, r_{AD0}, \beta]^T$ is then the initial state of the whole flight, and the four-dimensional guidance-parameter vector $\mathbf{p} = [w, \gamma, w_T, w_E]^T$ is the decision variable to be solved by shooting and by reinforcement-learning training. Single shooting is a direct transcription method for continuous-time optimal control. It is simple to implement and compatible with standard NLP solvers. Unlike collocation, which discretizes the state and the control simultaneously on a dense node set, single shooting parameterizes only the decision variables and recovers the trajectory by forward integration of the system dynamics, converting an infinite-dimensional optimal-control problem into a finite-dimensional parameter optimization problem.

On the engagement interval $[t_0, t_f]$, the guidance-parameter vector $\mathbf{p} = [w, \gamma, w_T, w_E]^T$ to be decided is represented by K piecewise-constant segments:

$$\mathbf{p}(t) = \mathbf{p}_k, \quad t \in [t_{k-1}, t_k), \quad k = 1, \ldots, K \tag{74}$$

where $K = 10$ and the segment boundaries are equally spaced. The guidance parameter $\mathbf{p}$ remains four-dimensional, but only weight ratios act in the closed-loop guidance law. To remove the common scale redundancy of the weights, the total evasion weight is fixed at $w_k = w = 10^4$. The actual decision variables of the shooting method are therefore $[\gamma_k, w_{T,k}, w_{E,k}]$ on each segment. The resulting expert parameters are further converted into the three-dimensional action labels defined by (78). The desired evasion distance is $d^* = 20$m.

Given a candidate parameter sequence $\{\mathbf{p}_k\}_{k=1}^K$, the engagement kinematics (1)–(5) are integrated forward by the fourth-order Runge–Kutta method (RK4) with step $\Delta t = 0.1$s:

$$X(t_{n+1}) = X(t_n) + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{75}$$

where X denotes the full system state and $k_1, \ldots, k_4$ are the RK4 increments computed from the closed-loop guidance law (49) under the parameters of the current segment. The NLP objective is constructed from the performance index (24):

$$\min_{\{\mathbf{p}_k\}_{k=1}^K} \mathcal{J} = s_w \sum_{i=1}^{2} \left[d_i^{\text{actual}} - d_i \right]^2 + s_{w_E} \int_{t_0}^{t_f} \mathbf{u}_A^T \mathbf{u}_A d\tau + s_{w_T} \| r_{AT}(t_{fT}) \|^2 \tag{76}$$

where $s_w = 2 \times 10^3$, $s_{w_E} = 55$, and $s_{w_T} = 22$ are fixed scaling coefficients, and d_i^{actual} is the actual minimum approach distance relative to NCV D_i. The resulting NLP is solved by IPOPT with an L-BFGS Hessian approximation.

The trajectories obtained by single shooting serve as the expert demonstrations for the subsequent BC-TD3 training. The expert data set is $\mathcal{D}_{\text{expert}} = \{\zeta^{(1)}, \zeta^{(2)}, \ldots, \zeta^{(N_{\text{expert}})}\}$, where each trajectory $\zeta^{(\ell)} = \{(\mathbf{s}_n^{(\ell)}, \mathbf{a}_n^{(\ell)})\}_{n=1}^{N_\ell}$ consists of N_ℓ state–action pairs extracted from the trajectory. Latin hypercube sampling draws $N_{\text{expert}} = 2400$ initial conditions in the 14-dimensional initial-state space, which yields about 8–15 $\times 10^4$ expert state–action samples.

5.2 Construction of the BC-TD3 network

5.2.1 State space, action space, and reward

State space (13-dimensional relative engagement state):

$$\mathbf{s}(t) = [\underbrace{r_{AD_1}, \dot{r}_{AD_1}, q_{y_1}, q_{z_1}}_{\text{NCV 1}}, \underbrace{r_{AD_2}, \dot{r}_{AD_2}, q_{y_2}, q_{z_2}}_{\text{NCV 2}}, \underbrace{r_{AT}, q_\theta, q_\psi}_{\text{Target}}, \underbrace{V_A, \bar{t}_{go}}_{\text{Evader}}]^T \in \mathbb{R}^{13} \tag{77}$$

where $\bar{t}_{go} = \min(t_{go,1}, t_{go,2})$ is the smaller time-to-go of the two NCVs. This state represents the relative geometry and kinematics of the evader with respect to the two NCVs and the target, and supports the guidance-parameter decision.

Action space (three-dimensional weight combination):

$$\mathbf{a}(t) = [\frac{w}{w_E}, \frac{w_T}{w_E}, \gamma]^T \in \mathbb{R}^3 \tag{78}$$

The guidance parameter $\mathbf{p} = [w, \gamma, w_T, w_E]^T$ remains four-dimensional. Because w is fixed, the three action components determine the weight relations required by the closed-loop guidance law. Writing $\mathbf{a} = [a_1, a_2, a_3]^T$ gives $w_E = w/a_1$, $w_T = a_2 w_E$, and $\gamma = a_3$. The meaning of each parameter is given at the end of Section 2.

The reward consists of a stepwise process reward during the evasion phase and a terminal reward settled after the phase-3 PN integral rollout. The process reward is

$$\begin{cases} R_1 = 0.05\log_{10}(\min(r_{AD1}, d^*) - 5), if\ r_{AD1,\min} \geqslant 6\text{m}\ else\ 0 \\ R_2 = 0.05\log_{10}(\min(r_{AD2}, d^*) - 5), if\ r_{AD2,\min} \geqslant 6\text{m}\ else\ 0 \end{cases} \tag{79}$$

where R_1 and R_2 are the process rewards for evading the two NCVs, and $r_{ADi,\min} = \min r_{ADi}(t)$ is the minimum approach distance to NCV D_i over the whole engagement.

The terminal reward is

$$\begin{cases} R_T = \dfrac{80}{5 + r_{AT,end}} \\ R_{bonus} = 100,\ if\ r_{AT,end} \leqslant 10\text{m}\ else\ 0 \\ R_{penalty} = -50,\ if\ \min(r_{AD1,\min}, r_{AD2,\min}) \leqslant 6\text{m}\ else\ 0 \end{cases} \tag{80}$$

where $r_{AT,end}$ is the terminal arrival error at the target obtained by the phase-3 PN rollout, and R_T is the arrival-accuracy reward based on that terminal accuracy. R_{bonus} is an additional reward given when evasion and target arrival are both completed, and $R_{penalty}$ is an additional penalty given when evasion fails.

5.2.2 Network architecture

The BC-TD3 framework uses a two-stage training paradigm: behavior-cloning (BC) pre-training, followed by TD3 reinforcement-learning fine-tuning with decaying BC regularization. Both stages are fully offline.

Actor network $\pi_\phi(\mathbf{s})$: it maps the 13-dimensional state to a 3-dimensional action,

$$\pi_\phi(\mathbf{s}) = \mathbf{a}_{\max} \cdot \tanh(\mathbf{W}_3\,\mathrm{ReLU}(\mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\mathbf{s} + \mathbf{o}_1) + \mathbf{o}_2) + \mathbf{o}_3) \tag{81}$$

The hidden layers are $[256, 256]$ with ReLU activation, and the output layer is tanh so that the action is bounded. Here $\mathbf{o}_1$, $\mathbf{o}_2$, and $\mathbf{o}_3$ are the bias vectors of the three layers.

Twin critic networks $Q_{\psi_1}(\mathbf{s}, \mathbf{a})$ and $Q_{\psi_2}(\mathbf{s}, \mathbf{a})$: each maps the concatenated pair $(\mathbf{s}, \mathbf{a})$ to a scalar Q-value,

$$Q_{\psi_j}(\mathbf{s}, \mathbf{a}) = \mathbf{W}_3^{(j)}\,\mathrm{ReLU}(\mathbf{W}_2^{(j)}\,\mathrm{ReLU}(\mathbf{W}_1^{(j)}[\mathbf{s}; \mathbf{a}] + \mathbf{o}_1^{(j)}) + \mathbf{o}_2^{(j)}) + o_3^{(j)}, \quad j = 1, 2 \tag{82}$$

The hidden layers are $[256, 256]$ with a linear output. The target networks $\pi_{\phi'}$, $Q_{\psi_1'}$, and $Q_{\psi_2'}$ are maintained by soft copies to stabilize training.

For BC pre-training, the actor is first trained by supervised behavior cloning on the expert data set ($\mathcal{D}_{\text{expert}} = \{(\mathbf{s}_n, \mathbf{a}_n)\}_{n=1}^{N_{\text{expert}}}$):

$$\mathcal{L}_{\text{BC}}(\phi) = \frac{1}{N}\sum_{n=1}^{N} \|\pi_\phi(\mathbf{s}_n) - \mathbf{a}_n\|^2 \tag{83}$$

The pre-trained actor parameters ϕ_{BC} provide a warm start for TD3 fine-tuning. In addition, the expert state–action pairs pre-fill the replay buffer $\mathcal{R}$, so that early TD3 training already benefits from high-quality demonstrations. The same expert data are sampled during fine-tuning to construct the BC regularization term in the actor loss.

The clipped double Q-learning and delayed policy updates of TD3 mitigate the Q-value overestimation and policy oscillation of DDPG, and therefore converge more steadily on multi-constraint tasks. Relative to DDPG, TD3 has three key mechanisms.

(1) Clipped double Q-learning. The target takes the minimum of the two critic estimates, which suppresses overestimation bias:

$$y_n = r_n + \gamma_{\mathrm{d}} \min_{j=1,2} Q_{\psi_{j'}}(\mathbf{s}_{n+1}, \tilde{\mathbf{a}}_{n+1}) \tag{84}$$

where the target action includes target-policy smoothing:

$$\tilde{\mathbf{a}}_{n+1} = \pi_{\phi'}(\mathbf{s}_{n+1}) + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathrm{clip}\left(\mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I}), -c, c\right) \tag{85}$$

The critic loss is

$$\mathcal{L}_{\mathrm{Critic}}(\psi_j) = \frac{1}{N} \sum_{n=1}^{N} (y_n - Q_{\psi_j}(\mathbf{s}_n, \mathbf{a}_n))^2, \quad j = 1, 2 \tag{86}$$

(2) Delayed policy updates with decaying BC regularization. The actor is updated once every d steps, rather than at every step, which reduces the coupling between a rapidly changing critic target and the policy update. At fine-tuning step k, BC-TD3 minimizes the following actor loss:

$$\begin{aligned} \mathcal{L}_{\mathrm{Actor}}(\phi; k) = & -\frac{1}{N} \sum_{n=1}^{N} Q_{\psi_1}(\mathbf{s}_n, \pi_\phi(\mathbf{s}_n)) + \lambda_{\mathrm{BC}}(k) \mathcal{L}_{\mathrm{BC}}(\phi), \\ \lambda_{\mathrm{BC}}(k) = & \lambda_{\mathrm{BC},0} \max\left(1 - \frac{k}{K_{\mathrm{BC}}}, 0\right). \end{aligned} \tag{87}$$

Here, the first term is the standard deterministic-policy objective and the second term constrains the actor toward the expert actions. In the implementation, $\lambda_{\mathrm{BC},0} = 1.0$ and $K_{\mathrm{BC}} = 200\,000$ fine-tuning steps. Thus, the expert constraint is strongest at the beginning of fine-tuning, decreases linearly to zero over the first 200 000 steps, and is removed thereafter so that the remaining updates are governed by the TD3 objective. For plain TD3, $\lambda_{\mathrm{BC}}(k) = 0$ throughout training.

(3) Soft updates of the target networks, executed at the same delayed frequency as the actor:

$$\phi' \leftarrow \tau\phi + (1-\tau)\phi', \qquad \psi_{j'} \leftarrow \tau\psi_j + (1-\tau)\psi_{j'} \tag{88}$$

where $\tau \ll 1$ (for example, $\tau = 0.005$).

Exploration noise in fine-tuning. During interaction with the environment, Gaussian noise is added to the actor output:

$$\mathbf{a}_n = \pi_\phi(\mathbf{s}_n) + \eta_n, \quad \eta_n \sim \mathcal{N}(\mathbf{0}, \sigma_e^2 \mathbf{I}) \tag{89}$$

where σ_e decays with the training episode, so that the policy moves from exploration toward exploitation.

The overall BC-TD3 network architecture is shown in Fig. 2.

The training hyperparameters of BC-TD3 and TD3 are listed in Table 3.

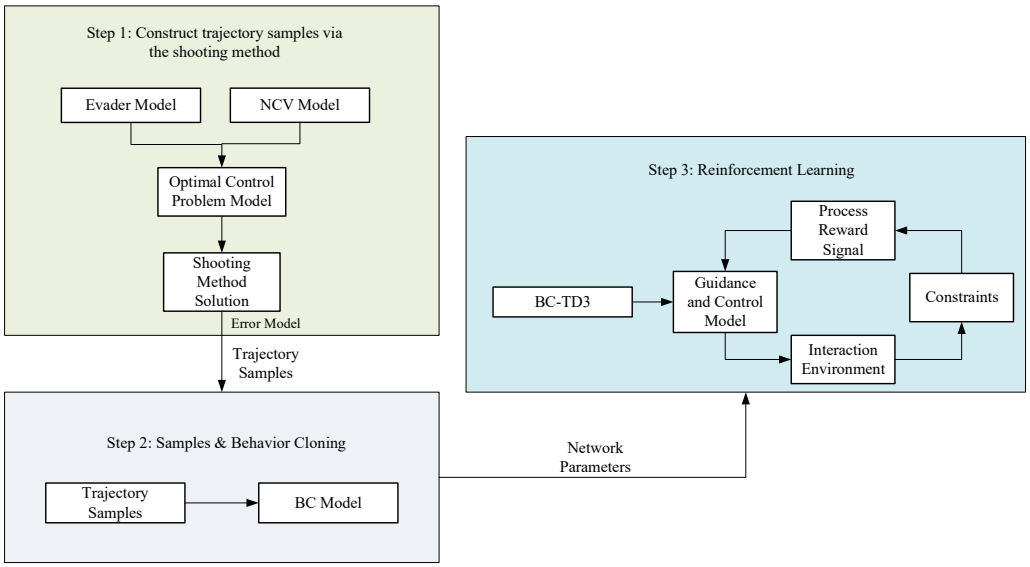


Figure 2: BC-TD3 network architecture.

Table 3: Training hyperparameters of TD3 and BC-TD3.

Hyperparameter	TD3	BC-TD3
Total training steps	300 000	300 000
Learning rate	1×10^{-4}	1×10^{-4}
Batch size	256	256
Discount factor γ_{d}	0.99	0.99
Soft-update coefficient τ	0.005	0.005
Replay capacity	300 000	300 000
Steps before updates	1 000	0
Training frequency	every step	every step
Policy delay	3	3
Network structure	[256, 256]	[256, 256]
Exploration noise σ	0.1	0.1
Target-policy noise	0.2	0.2
Target-noise clip	0.5	0.5
Gradient clip	±10	±10
BC pre-fill ratio	—	0.33
Initial BC regularization $\lambda_{\mathrm{BC},0}$	—	1.0
BC regularization decay horizon K_{BC}	—	200 000 steps

5.2.3 Training of the BC-TD3 network

The complete offline training procedure is as follows.

Algorithm 1 Offline BC-TD3 training for guidance-parameter optimization

Input: expert data set $\mathcal{D}_{\text{expert}}$, engagement simulator Env
Output: trained actor π_ϕ
// Stage 1: BC pre-training
Initialize the actor π_ϕ with random weights
for epoch $= 1$ to E_{BC} do
  Sample a minibatch $\{(\boldsymbol{s}_n, \boldsymbol{a}_n)\}$ from $\mathcal{D}_{\text{expert}}$
  Update ϕ by minimizing $\mathcal{L}_{\text{BC}}$ (83)
end for
Pre-fill the replay buffer $\mathcal{R}$ with $\mathcal{D}_{\text{expert}}$
// Stage 2: TD3 fine-tuning
Initialize the critics ψ_1, ψ_2 and the target networks $\phi' \leftarrow \phi$, $\psi'_j \leftarrow \psi_j$
for episode $= 1$ to E_{TD3} do
  Reset Env to a random initial condition $\boldsymbol{s}_0$
  for $n = 0, 1, 2, \ldots$ do
    $\boldsymbol{a}_n = \pi_\phi(\boldsymbol{s}_n) + \boldsymbol{\eta}_n$ // 89
    Convert $\boldsymbol{a}_n$ to $\boldsymbol{p}_n$, obtain $\boldsymbol{u}_A^*(t)$ from (49), and apply magnitude saturation
    Observe $\boldsymbol{s}_{n+1}, r_n$ (79–80)
    Store $(\boldsymbol{s}_n, \boldsymbol{a}_n, r_n, \boldsymbol{s}_{n+1})$ in $\mathcal{R}$
    Sample a minibatch of N transitions from $\mathcal{R}$
    Compute the target y_n (84–85)
    Update the critics ψ_1, ψ_2 by minimizing $\mathcal{L}_{\text{Critic}}$ (86)
    if $n \bmod d = 0$ then // delayed update
      Sample expert pairs, compute $\lambda_{\text{BC}}(k)$, and update ϕ by minimizing $\mathcal{L}_{\text{Actor}}$ (87)
      Soft-update the target networks ϕ', ψ'_j (88)
    end if
  end for
end for
// Deployment (online inference)
return π_ϕ — at each guidance cycle, $\boldsymbol{a} = \pi_\phi(\boldsymbol{s})$, convert it to $\boldsymbol{p}$, and pass it with the identified matrices to (49)

6 Simulation Results

6.1 Simulation conditions

This section evaluates the proposed integrated optimal evasion and guidance law and the BC-TD3 parameter optimization in a three-dimensional engagement between an evader and two NCVs. The study has three parts. Section 6.1 summarizes the simulation conditions, including the engagement geometry, the vehicle models, the training settings, and the success criteria. Section 6.2 examines evasion and guidance without noise, with emphasis on learning efficiency, representative trajectories, and Monte Carlo success rates, and compares classical and learning-based baselines. Section 6.3 further evaluates closed-loop robustness under measurement noise, and compares BC-TD3 with and without online parameter identification, so as to quantify the benefit of the sparse FF-RLS module developed in Section 4.

All simulations are three-dimensional four-body engagements: the evader reaches a fixed target while avoiding two NCVs. The NCVs use the classical proportional navigation in Assumption 1. The evader acceleration command is generated by the closed-loop guidance law (49). The guidance parameter $\mathbf{p} = [w, \gamma, w_T, w_E]^T$ remains four-dimensional. The trained BC-TD3 model outputs online the three-dimensional action in (78), from which $\mathbf{p}$ is determined.

Unless otherwise stated, the desired evasion distance is $d^* = 20\,\text{m}$, the NCV lethal radius is 6 m, the successful target-arrival threshold is 10 m, and the guidance period is $\Delta t = 0.1\,\text{s}$. In the simulation, the maximum normal accelerations available to the evader and the NCVs are $8g$ and $10g$, respectively. A command that exceeds this magnitude is applied at the corresponding limit. All compared methods use the same setting.

Two complementary evaluation settings are considered. In the noise-free setting (Section 6.2), relative range, line-of-sight angles and rates, and attitude angles are assumed exact, so the matrix elements in (49) can be obtained analytically from (13) and (15). In the noisy setting (Section 6.3), zero-mean Gaussian white noise is added to both measurement classes as in Assumption 2. The standard deviations of the angular measurements (line-of-sight angle, line-of-sight rate, flight-path angle, and heading angle) are $\sigma_q = \sigma_\theta = \sigma_\psi = 3^\circ$ and $\sigma_{\dot{q}} = 0.03\text{rad/s}$. The standard deviations of the range measurements are $\sigma_r = 5\,\text{m}$ and $\sigma_{\dot{r}} = 0.5\,\text{m/s}$, comparable to the typical accuracy of radar-seeker ranging and coherent velocity measurement. The constant-acceleration Kalman filter (59)–(62) uses a jerk intensity $\sigma_j = 20\,\text{m/s}^3/\sqrt{\text{Hz}}$ and $\sigma_{a0} = 18g$ in the initial covariance. The sparse FF-RLS identifier then recovers the required parameters from the relative kinematics.

Besides the network structure and the common TD3 hyperparameters already given in Section 5, the remaining vehicle, numerical, baseline, and identifier parameters needed to reproduce the experiments are summarized in Table 4.

Table 4: Supplementary simulation and implementation parameters.

Parameter	Symbol	Value or implementation
Evader speed	V_A	$500\,\text{m/s}$
Speed of both NCVs	$V_{D1} = V_{D2} = V_D$	$600\,\text{m/s}$
NCV proportional-navigation gain	K_D	4
Dynamic integration step	h	0.1 s (RK4)
Guidance and policy update period	Δt	0.1 s
Maximum simulation time	$t_{\max}$	
Termination events	—	
Closest-approach detection	—	
Sine-maneuver start time	t_0	
Sine-maneuver amplitude and frequency	A_ω, f	$54\,\text{m/s}^2,\ 0.15\,\text{Hz}$
Policy action bounds	a_1, a_2, a_3	
BC pre-training epochs	E_{BC}	
Exploration noise and its decay	σ_e	0.1 (initial value)
Training random seeds	—	
Sparse FF-RLS parameters	ρ, η, δ	
Normalization scales	r_0, a_0	

The dynamic integration step h is used to integrate the state equations, and Δt is the update period of the guidance law and the policy. In the subsequent representative trajectories and Monte Carlo trials, every method should use the same integration, termination, and closest-approach detection settings. In particular, the minimum distance near the 6 m evasion threshold should be determined by a sufficiently small h or by inter-step event or interpolation detection, so that time discretization does not change the success decision.

Representative-trajectory performance is examined first. The subsequent Monte Carlo trials sample the multi-dimensional initial-condition space by Latin hypercube sampling (LHS) and, with the same random seeds for all compared methods, report the evasion, target-arrival, and joint success rates.

The initial engagement conditions of the representative trajectories are given in Table 5.

Table 5: Initial engagement conditions and scenario parameters of the representative trajectories.

Object	Initial position/m	Initial flight-path angle	Initial heading angle
Evader A	(10000,6000,3500)	$\theta_{A0} = -20^\circ$	$\psi_{V_A0} = 155^\circ$
NCV D_1	(1000,0,0)	$\theta_{D10} = 40^\circ$	$\psi_{V_{D1}0} = -20^\circ$
NCV D_2	(-1000,0,0)	$\theta_{D20} = 40^\circ$	$\psi_{V_{D2}0} = -20^\circ$

To verify the proposed algorithm, four guidance methods are compared: (1) proportional navigation with $K_A = 4$; (2) sinusoidal evasive maneuvering, with acceleration commands $u_{Ay} = A_\omega \cos(2\pi f(t - t_0))$ and $u_{Az} = A_\omega \sin(2\pi f(t - t_0))$, where f is the frequency and t_0 is the evasion

start time, and $A_\omega = 54\mathrm{m/s}^2$, $f = 0.15\mathrm{Hz}$; (3) the optimal integrated evasion and guidance law based on TD3; and (4) the optimal integrated evasion and guidance law based on BC-TD3, which is the method proposed in this paper.

The Monte Carlo parameter distributions and ranges are given in Table 6.

Table 6: Monte Carlo parameter distributions and ranges.

Parameter	Range	Distribution
q_{yi}	$[20°, 70°]$	Uniform
q_{zi}	$[-20°, 20°]$	Uniform
η_{yAi0}, η_{yDi0}	$[-15°, 15°]$	Normal $N(0, 6^2)$, uniform
η_{zAi0}, η_{zDi0}	$[-10°, 10°]$	Normal $N(0, 6^2)$, uniform
r_{AD0}	$[10.5, 13.5]$km	Uniform
β	$[-1.5, 1.5]$km	Uniform

6.2 Evasion and guidance without measurement noise

This subsection verifies the proposed BC-TD3 evasion and target-arrival guidance law under ideal (noise-free) information. Because the guidance matrices can be computed exactly, performance differences among the methods are attributed mainly to the quality of the online guidance-parameter policy, rather than to sensor or identification error. Three complementary studies are reported: (i) the learning curves of TD3 and BC-TD3; (ii) the three-dimensional engagement trajectories of each evasion method, initialized as in Table 5, together with enlarged views of the encounters with each NCV; and (iii) Monte Carlo success rates of each method based on Table 6.

The learning curves of TD3 and BC-TD3 are shown in Fig. 3.

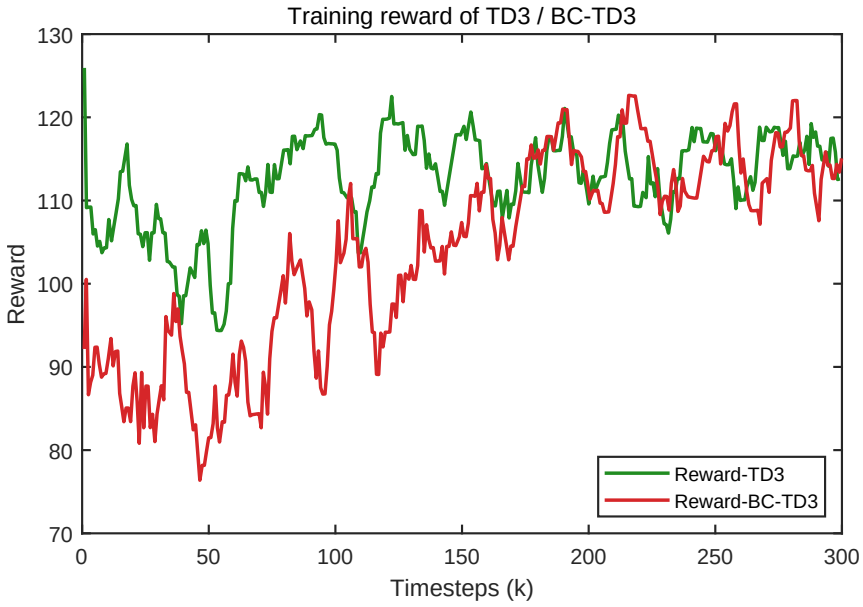


Figure 3: Training reward curves of BC-TD3 and TD3.

The three-dimensional engagement trajectories of BC-TD3 and TD3 are shown in Fig. 4 and Fig. 5.

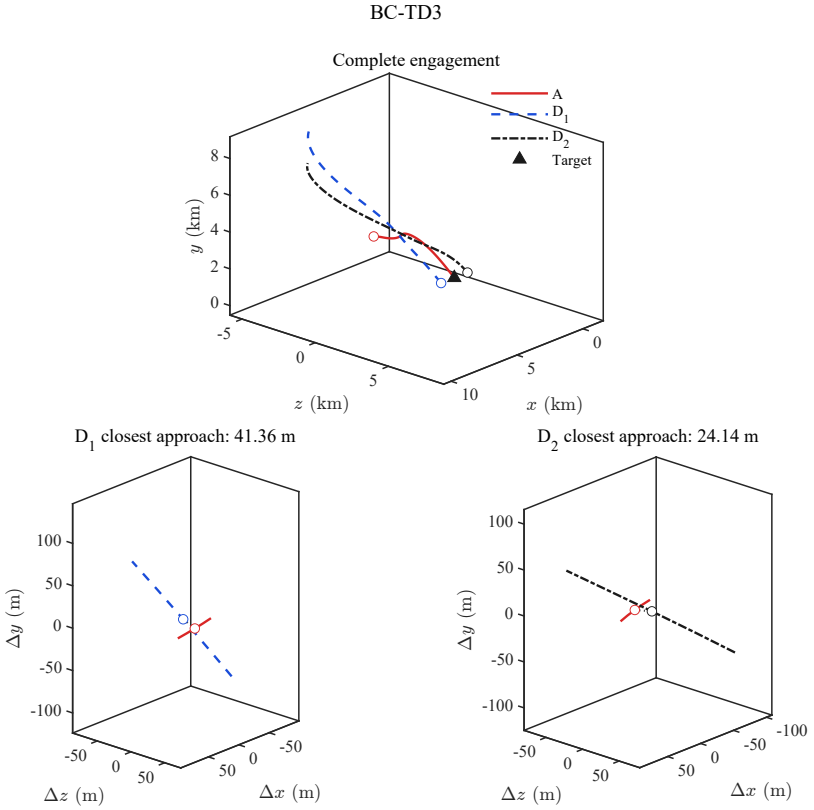


Figure 4: Three-dimensional BC-TD3 trajectory and enlarged views of the NCV encounters.

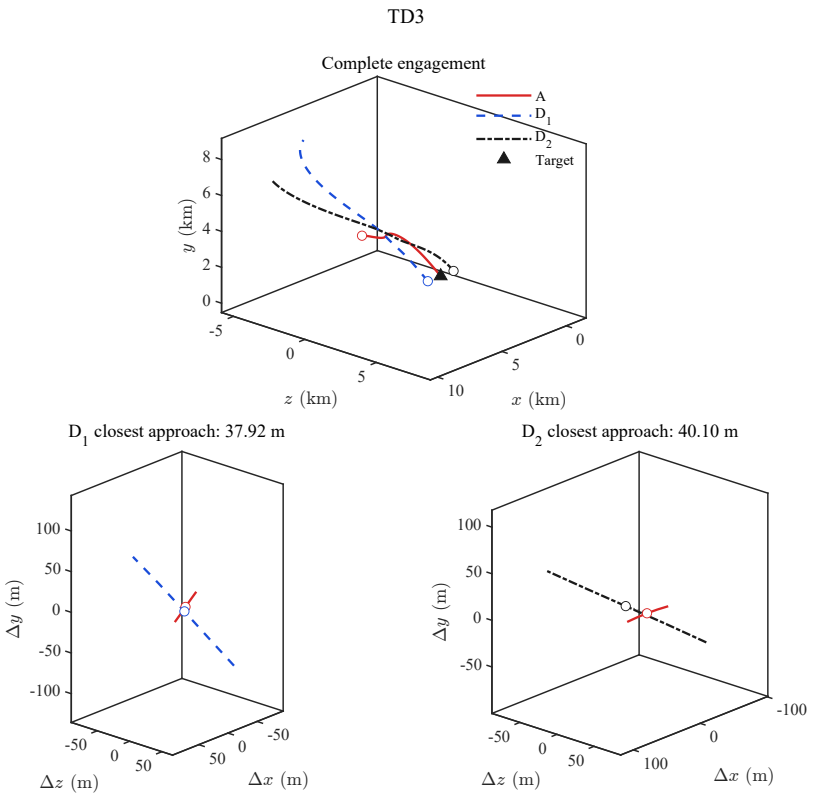


Figure 5: Three-dimensional TD3 trajectory and enlarged views of the NCV encounters.

The three-dimensional engagement trajectory of the sinusoidal maneuver is shown in Fig. 6.

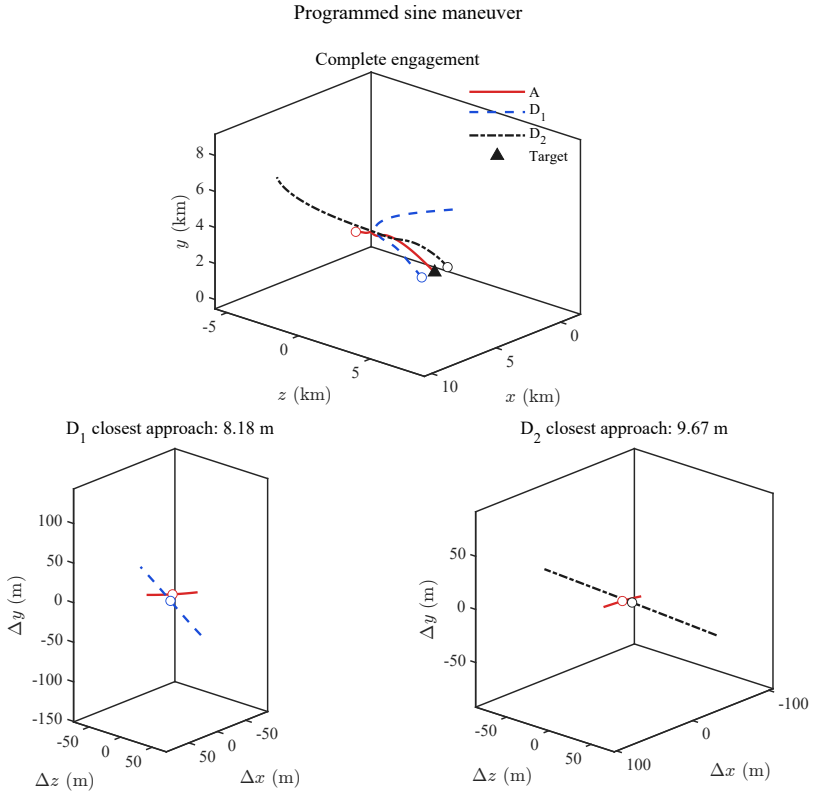


Figure 6: Three-dimensional trajectory of the sinusoidal maneuver and enlarged views of the NCV encounters.

The acceleration histories of the four guidance methods are shown in Fig. 7.

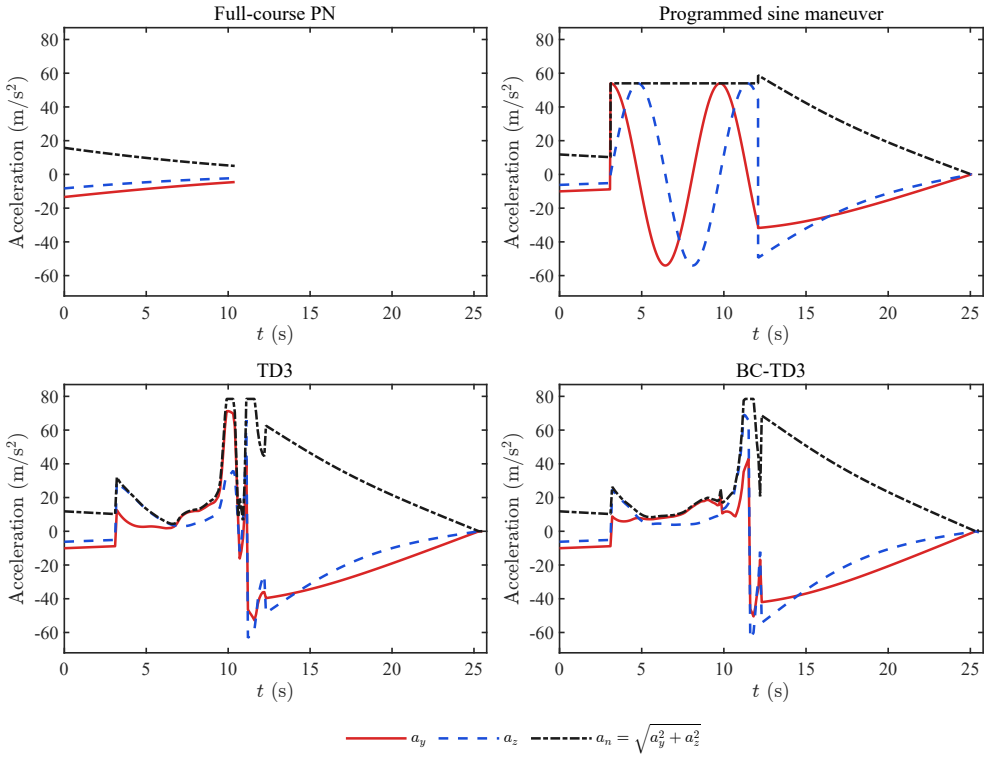


Figure 7: Normal acceleration histories of the compared methods.

The histories of r_{AD1}, r_{AD2}, and r_{AT} for the four guidance methods are shown in Fig. 8.

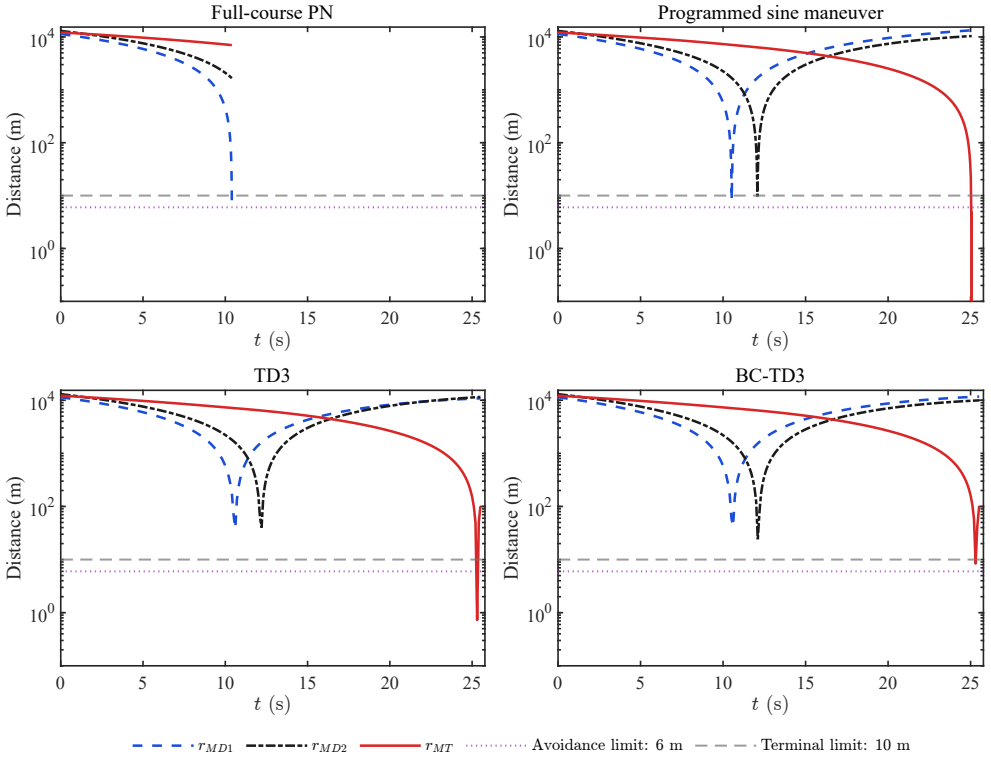


Figure 8: Range histories between the evader and the two NCVs and the target.

From Fig. 7–8, under full-course proportional navigation the evader flies nearly level with a small load factor. At about 10.5 s, r_{AD1} reaches the evasion threshold (6 m), evasion fails, the simulation terminates early, and target-arrival guidance is not completed. The programmed sinusoidal maneuver applies a sustained large sinusoidal load factor in the middle phase, and a_n remains high for a long time. The minimum values of r_{AD1} and r_{AD2} are both slightly above 6 m, and r_{AT} falls below 10 m in the terminal phase, so evasion and target arrival can both be completed, but the safety margin relative to the NCVs is small. TD3 and BC-TD3 adaptively increase the maneuver in the middle phase. Near 10–12 s, a_n reaches the $8g$ magnitude limit and then the load factor gradually decreases into terminal guidance. For both methods, the minimum values of r_{AD1} and r_{AD2} are clearly larger than those of the programmed maneuver, and the terminal r_{AT} is below 10 m, so both complete dual-NCV evasion and target-arrival guidance. The maneuver of BC-TD3 is smoother and more stable.

To compare performance, 300 Monte Carlo simulations were run for BC+TD3, TD3, proportional navigation, and the programmed evasive maneuver. The comparison is given in Table 7.

Table 7: Algorithm performance comparison (without noise).

Method	Dual-NCV evasion success rate $\min(r_{AD1}, r_{AD2}) \geqslant 6$m	Target-arrival success rate (evasion success and $r_{AT,end} \leqslant 10$m)
Proportional navigation to the target only	0.0%	0.0%
Programmed maneuver (sinusoidal)	76.3%	74.3%
TD3	93.7%	93.0%
BC+TD3	94.7%	94.3%

Without a maneuver, or with only a fixed programmed maneuver, the evader has difficulty completing dual-NCV evasion and target arrival at the same time. Under proportional navigation to the target only, both success rates are 0.0%. The programmed sinusoidal maneuver raises the two indices to 76.3% and 74.3%, which is still clearly insufficient. TD3 and BC+TD3 both complete the

evasion and target-arrival guidance tasks well: the evasion and target-arrival success rates of TD3 are 93.7% and 93.0%, and BC+TD3 further reaches 94.7% and 94.3%. BC+TD3 is higher than TD3 on both success rates, and its overall performance and engineering applicability are better.

6.3 Evasion and guidance under measurement noise

In a practical engagement, the matrix elements required by the guidance law (49) cannot be obtained from noise-free kinematics. As in Assumption 2, onboard line-of-sight angles, line-of-sight rates, and attitude angles are corrupted by Gaussian white noise, while relative range and range rate are corrupted by Gaussian white noise of a relatively smaller magnitude. Substituting the angular measurements into (13) and (15) introduces biased and mutually correlated matrix errors. This subsection therefore repeats the engagement study under sensor noise and examines whether the sparse FF-RLS identifier of Section 4 can recover closed-loop performance. Two results are given: (i) a typical noisy BC-TD3 trajectory with online identification and enlarged encounter views; and (ii) a Monte Carlo comparison of five policies, namely BC-TD3 without and with identification, TD3, the sinusoidal maneuver, and PN.

Fig. 9 shows a typical noisy flight trajectory of BC-TD3 combined with online sparse FF-RLS identification. The initial condition is based on Table 5, and the noise intensity is the same as in Section 6.1, namely $\sigma_q = \sigma_\theta = \sigma_\psi = 3^\circ$, $\sigma_{\dot{q}} = 0.03\text{rad/s}$, $\sigma_r = 5\,\text{m}$, and $\sigma_{\dot{r}} = 0.5\,\text{m/s}$.

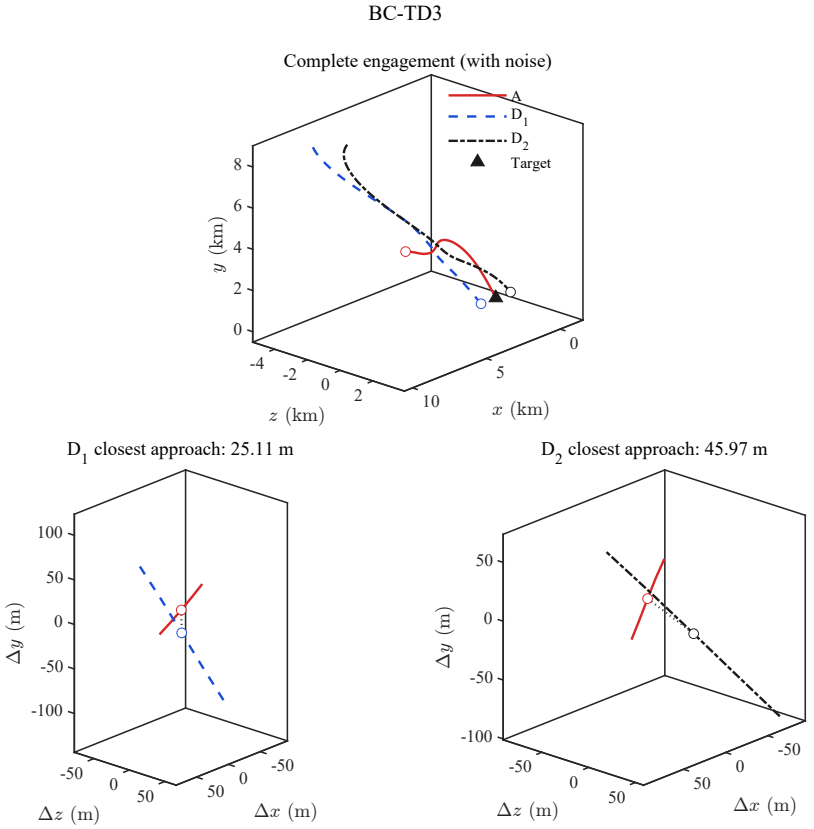


Figure 9: Three-dimensional trajectory of BC-TD3 with identification and enlarged encounter views (with measurement noise).

A typical noisy flight trajectory without online sparse FF-RLS identification is shown in Fig. 10.

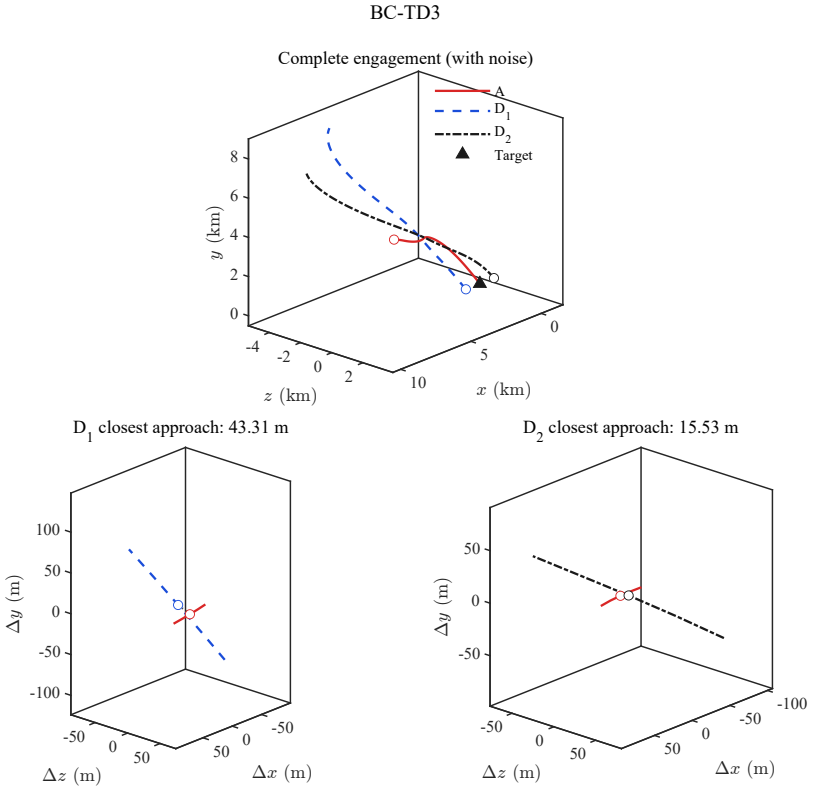


Figure 10: Three-dimensional trajectory of BC-TD3 without identification and enlarged encounter views (with measurement noise).

From Fig. 9–10, after the measurement noise above is introduced and processed by online sparse FF-RLS identification, the overall guidance and flight are almost unaffected: the evader can still complete a large maneuver, arrive at the target accurately, and maintain effective evasion. Without identification, the measurement noise enters the guidance loop directly. The closest evasion distances are 43.31 m and 15.53 m. The evasion distance from the second NCV D_2 decreases markedly, and the terminal target-arrival accuracy is also affected to some extent. Thus, without online sparse FF-RLS identification, both evasion and target-arrival performance degrade, whereas identification effectively suppresses the disturbance of measurement noise on evasion and target-arrival guidance.

The Monte Carlo results under measurement noise are given in Table 8.

Table 8: Algorithm performance comparison (with noise).

Method (without identification unless noted)	Dual-NCV evasion success rate $\min(r_{AD1}, r_{AD2}) \geqslant 6$m	Target-arrival success rate (evasion success and $r_{AT,end} \leqslant 10$m)
Proportional navigation to the target only	0.0%	0.0%
Programmed maneuver (sinusoidal)	76.3%	74.3%
TD3	92.3%	91.7%
BC+TD3	93.7%	93.3%
BC+TD3 with identification	96.7%	96.0%

Under measurement noise, the evader still has difficulty completing dual-NCV evasion and target arrival at the same time if it does not maneuver or uses only a fixed programmed maneuver. Under proportional navigation to the target only, both success rates are 0.0%. The programmed sinusoidal maneuver raises the two indices to 76.3% and 74.3%, which is still clearly insufficient. The evasion and target-arrival success rates of TD3 and BC+TD3 are 92.3% and 91.7%, and 93.7% and 93.3%, respectively. They can still complete the task, but both are slightly lower than in the noise-free case.

After online sparse FF-RLS parameter identification is introduced, BC+TD3 with identification further raises the two success rates to 96.7% and 96.0%, which is markedly higher than BC+TD3 without identification. This indicates that parameter identification can suppress measurement-noise disturbance and improve the robustness of evasion and target-arrival guidance.

7 Conclusion

This paper addresses the requirement that an evader simultaneously avoid two NCVs and retain terminal accuracy at a specified target, and proposes a three-dimensional integrated evasion and guidance framework. A four-body relative-motion model is established to describe the coupled interaction among the evader, the target, and the two NCVs. On this basis, the dual-NCV evasion requirement, the target-arrival accuracy, and the control-effort penalty are placed in a unified performance index, and an optimal closed-loop guidance law is derived. The guidance law explicitly accounts for cross-coupling: one maneuver simultaneously affects both evasive encounters and terminal arrival at the target.

To select guidance weights adaptively under different engagement geometries, a BC-TD3 parameter-optimization strategy is proposed. Expert demonstrations generated by single shooting are used to pre-train the actor and pre-fill the replay buffer. During TD3 fine-tuning, a gradually decayed behavior-cloning regularization term retains the expert constraint in the early stage and then transfers the actor toward reward-driven optimization, thereby improving its adaptability in a nonlinear multi-constraint environment. In addition, a recursive least-squares identifier with sparse regularization and a forgetting factor estimates the nontrivial parameters of the state-space model from noisy measurements, which reduces the sensitivity of the guidance loop to measurement errors.

Simulation results show that the proposed method can reliably complete evasion and target arrival at the same time in a three-dimensional dual-NCV scenario. In noise-free Monte Carlo trials, the evasion success rate and the target-arrival success rate of BC-TD3 are 94.7% and 94.3%, respectively, which are higher than those of classical proportional navigation, programmed sinusoidal maneuvering, and TD3 without behavior cloning. Under measurement noise, introducing sparse online identification further raises the two success rates to 96.7% and 96.0%. These results indicate that the proposed framework can provide an effective and robust solution for integrated multi-NCV evasion and terminal guidance. Future work will further consider vehicle dynamics closer to practice, heterogeneous NCVs, actuator constraints, and verification under a wider range of uncertainties.

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